MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

FLIP-FLOP, D TYPE, DIGITAL, KEY CODE KDR

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic D type flip-flop module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.19 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be KDR.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/61-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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4.4.2 Order of testing. Test conditions A through Z shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Z in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through W and Z shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition Z, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests E through W and Z shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through z shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Military Specification MIL-M-28787/61 supersedes Department of the Navy, Navy Ordnance Systems Command Drawing No. 10001-2658775. Part number M28787/61-1 supersedes part numbers 2658775 and M28787/61.

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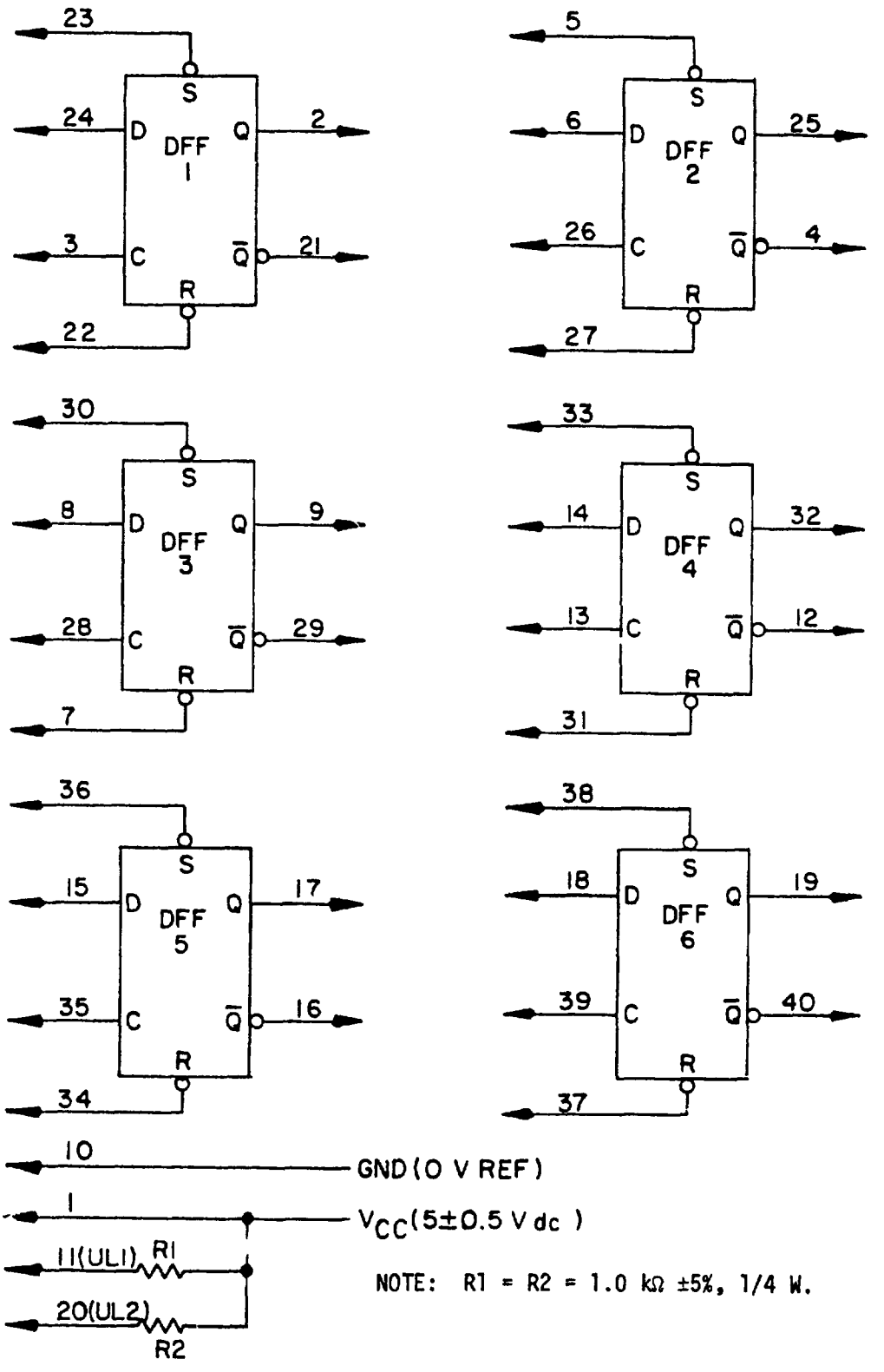


FIGURE 1. KDR module function diagram, D type flip-flop.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A12	V_{OUT} (low loaded)	III	S	10	800	10	800	10	800	10	800	mV
B1-B18	Input breakdown current	IV	S	0	1	0	1	0	1	0	1	mA
B19-B24	Input breakdown current	IV	S	0.00	1.25	0.00	1.25	0.00	1.25	0.00	1.25	mA
C	Isolation	V	S	10	∞	1	∞	10	∞	1	∞	MΩ
D1-D24	Input capacitance	VI	Q	2	18	2	18	2	18	2	18	pF
E1-E54	Output voltage, low level	VII	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
F1-F78	Output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
G1-G6	Input current, low level	VIII	X	0.40	1.73	0.40	1.73	0.40	1.73	0.40	1.73	mA
G7-G18	Input current, low level	VIII	X	0.80	3.46	0.80	3.46	0.80	3.46	0.80	3.46	mA
G19-G24	Input current, low level	VIII	X	1.20	5.19	1.20	5.19	1.20	5.19	1.20	5.19	mA
H1-H6	Input current, high level	IX	X	0	30	0	40	0	30	0	40	µA
H7-H18	Input current, high level	IX	X	0	60	0	80	0	60	0	80	µA
H19-H24	Input current, high level	IX	X	0	90	0	120	0	90	0	120	µA
J1-J12	Output short circuit current	X	X	20	55	20	55	20	55	20	55	mA
J13,J14	Output short circuit current (UL1,UL2)	X	X	5.1	5.9	5.1	5.9	5.1	5.9	5.1	5.9	mA
K1	Power supply current, low level	XI	S	1	90	1	90	1	90	1	90	mA
K2	Power supply current, high level	XI	S	1	90	1	90	1	90	1	90	mA

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
L1-L12	Transition time (t_{TLH})	XII	S	4	45	4	45	4	50	4	50	ns
M1-M12	Transition time (t_{THL})	XII	S	4	20	4	20	4	25	4	25	ns
N1-N12	Propagation delay time (t_{PHL})	XIII	S	6	50	6	50	6	52	6	52	ns
P1-P12	Propagation delay time (t_{PLH})	XIII	S	6	30	6	30	6	35	6	35	ns
R1-R6	Setup time											
	TA	XIV	S	21	73	21	73	21	73	21	73	ns
	TB	XIV	S	431	591	431	591	431	591	431	591	ns
	TC	XIV	S	900	1100	900	1100	900	1100	900	1100	ns
R7-R12	Setup time											
	TA	XIV	S	471	623	471	623	471	623	471	623	ns
	TB	XIV	S	431	591	431	591	431	591	431	591	ns
	TC	XIV	S	900	1100	900	1100	900	1100	900	1100	ns
S1-S6	Hold time											
	TA	XIV	S	450	598	450	598	450	598	450	598	ns
	TB	XIV	S	431	591	431	591	431	591	431	591	ns
	TC	XIV	S	900	1100	900	1100	900	1100	900	1100	ns
S7-S12	Hold time											
	TA	XIV	S	0	48	0	48	0	48	0	48	ns
	TB	XIV	S	431	591	431	591	431	591	431	591	ns
	TC	XIV	S	900	1100	900	1100	900	1100	900	1100	ns
T1-T6	Maximum clock frequency/minimum clock pulse width											
	TA	XV	S	60	115	60	115	60	115	60	115	ns
	TB	XV	S	73	121	73	121	73	121	73	121	ns
	TC	XV	S	160	176	160	176	160	176	160	176	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
T7-T12	Maximum clock frequency/minimum clock pulse width											
	TA	XV	S	140	203	140	203	140	203	140	203	ns
	TB	XV	S	73	121	73	121	73	121	73	121	ns
	TC	XV	S	160	176	160	176	160	176	160	176	ns
U1-U6	Threshold clock											
	TB	XVI	S	440	585	440	585	440	585	440	585	ns
	TC	XVI	S	900	1100	900	1100	900	1100	900	1100	ns
V1-V12	Propagation delay time (t_{PHL})	XVII	S	6	45	6	45	6	50	6	50	ns
W1-W12	Propagation delay time (t_{PLH})	XVIII	S	6	25	6	25	6	30	6	30	ns
X1-X24	Output crosstalk, low level and high level	XIX	Q	0	400	0	400	0	400	0	400	mV
Y	Power supply transient amplitude	XX	Q	0	400	0	400	0	400	0	400	mV
Z	Power supply transient rise time	XXI	S	1	2	1	2	1	2	1	2	ms

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Supply voltage	5 V dc ±10%	
Maximum power dissipation	0.5 W (outputs not loaded)	
Speed (clock frequency)	DC to 12 MHz	
Input threshold voltage		
Low level	0.8 V maximum	
High level	2.0 V minimum	
Input current, low level	(V_{IL} = 0 V)	(V_{IL} = 0.4 V)
Data inputs	-1.73 mA maximum	-1.6 mA maximum
Clock and set inputs	-3.46 mA maximum	-3.2 mA maximum
Reset inputs	-5.19 mA maximum	-3.2 mA maximum
Input current, high level	(V_{IH} = 2.4 V)	(V_{IH} = 5.5 V)
Data inputs	40 µA maximum	1 mA maximum
Clock and set inputs	80 µA maximum	1 mA maximum
Reset inputs	120 µA maximum	1.25 mA maximum
Output voltage		
Low level (I_{OL} = 16 mA)	0.4 V maximum	
High level (I_{OH} = -400 µA)	2.4 V minimum	
Output drive		
Low level (V_{OL} = 0.4 V)	16 mA maximum	
High level (V_{OH} = 2.4 V)	-400 µA maximum	
Pulse width		
Clock	42 ns minimum	
Set	40 ns minimum	
Reset	40 ns minimum	
Propagation delay time (clock-to-output)		
t_{PHL}	6 ns minimum, 52 ns maximum	
t_{PLH}	6 ns minimum, 35 ns maximum	
Propagation delay time (set-to-output)		
t_{PHL}	6 ns minimum, 50 ns maximum	
t_{PLH}	6 ns minimum, 30 ns maximum	
Propagation delay time (reset-to-output)		
t_{PHL}	6 ns minimum, 50 ns maximum	
t_{PLH}	6 ns minimum, 30 ns maximum	
Setup time (t_{SETUP}) (data-to-clock)	20 ns minimum	
Hold time (t_{HOLD}) (data-to-clock)	5 ns minimum	
Clock triggering edge	Positive	

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* 6.5 Module function. This module contains six D type flip-flops that have independent data, clock, set, and reset inputs, and complementary data outputs.

6.5.1 Input to output relationship. Each D type flip-flop has the input to output relationship of figure 2 and table II for positive logic.

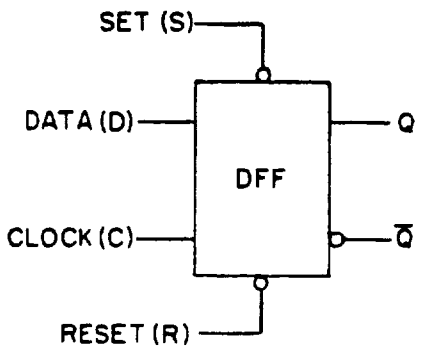


FIGURE 2 Logic diagram of functional circuits.

TABLE II. Input to output relationship. 1/ - 3/

Input t_n			Output t_{n+1}		
S	R	D	Q	$\overline{Q}$	Operation
H	H	L	L	H	Data entry
H	H	H	H	L	Data entry
H	L	X	L	H	Reset
L	H	X	H	L	Set
L	L	X	H	H	- - -

1/ X - Don't care.
2/ t_n - Time period before positive clock transition.
3/ t_{n+1} - Time period after positive clock transition.

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6.5.1.1 Operational description. This is an edge triggered D type flip-flop with independent set and reset inputs and complementary $\overline{Q}$ and Q outputs. Input information on the data input is transferred to the output on the positive edge of the clock pulse. Clock triggering occurs at a voltage level during the rise transition of the clock pulse. After the clock input threshold voltage has been passed, the data input is locked out. The set and reset inputs are independent of the clock. A low level input to the set line sets Q to a high level, and a low level input to the reset line sets Q to a low level. Simultaneous application of low level to the set and reset inputs will force both outputs to high level.

6.5.1.2 Setup time (t_{setup}) and hold time (t_{hold}). The logic level beiny applied to the data input must be present for a minimum of 20 ns before the clock input triggering level is reached and must be maintained for at least 5 ns after the clock pulse triggering level has been applied.

6.5.1.3 Up-level source (UL1, UL2). Two independent 1.0 kΩ resistors are provided on the module for a convenient up-level source for unused inputs.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

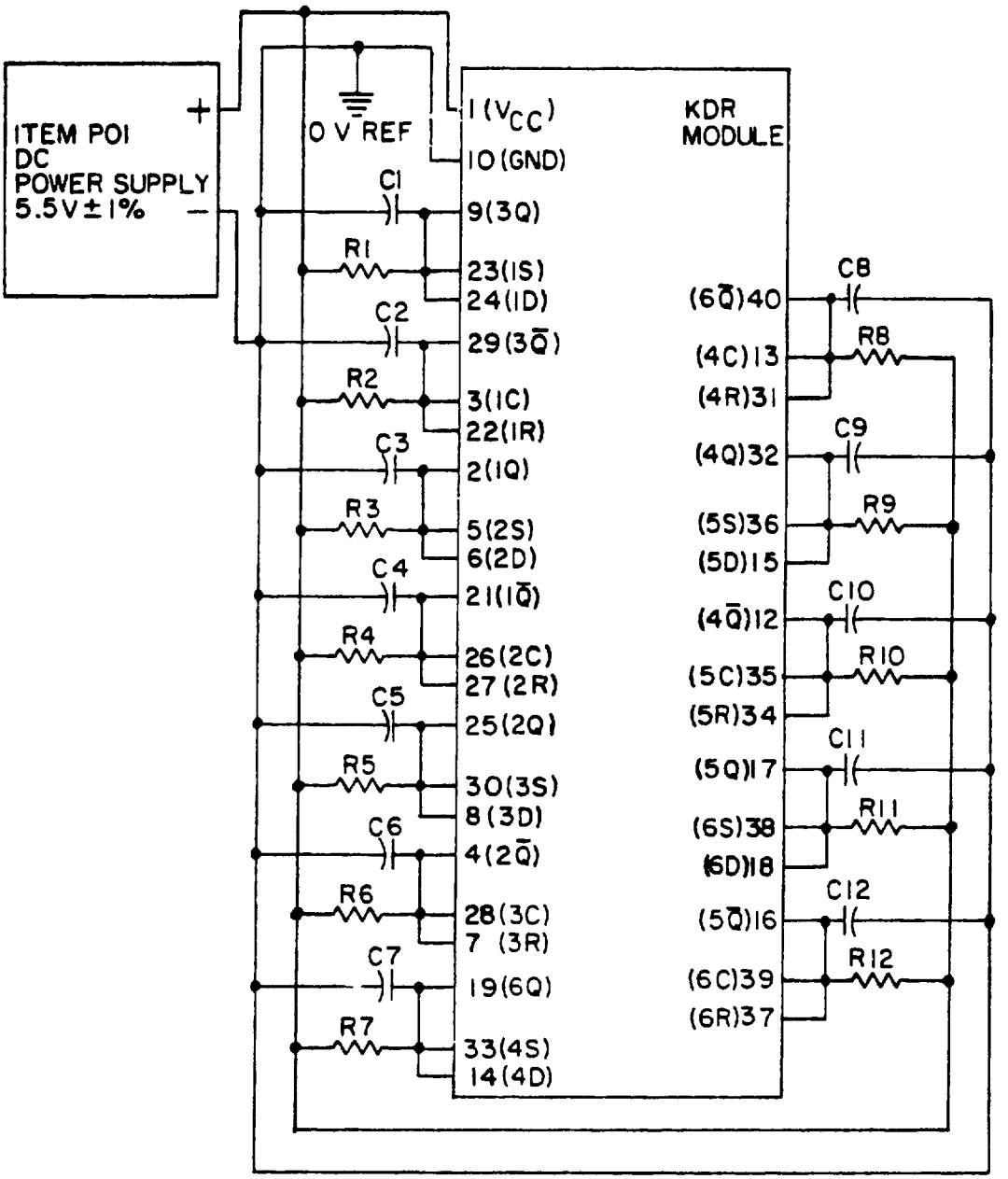
6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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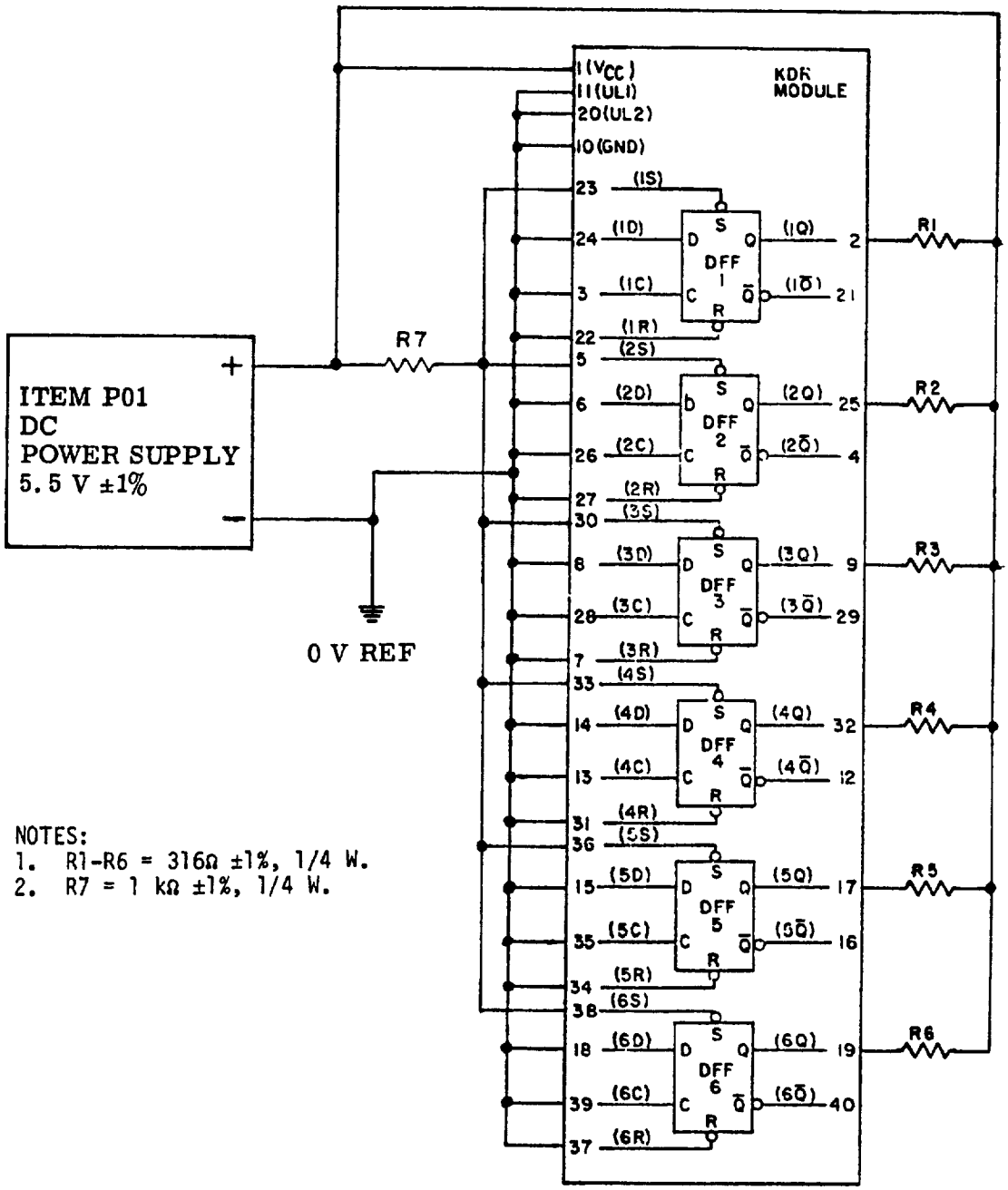
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NOTES:
1. R1-R12 = 316Ω ±1%, 1/4 W.
2. C1-C12 = 100 pF ±20%.

FIGURE 3. Operating temperature and life test circuit.

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* FIGURE 4. Component temperature test circuit.

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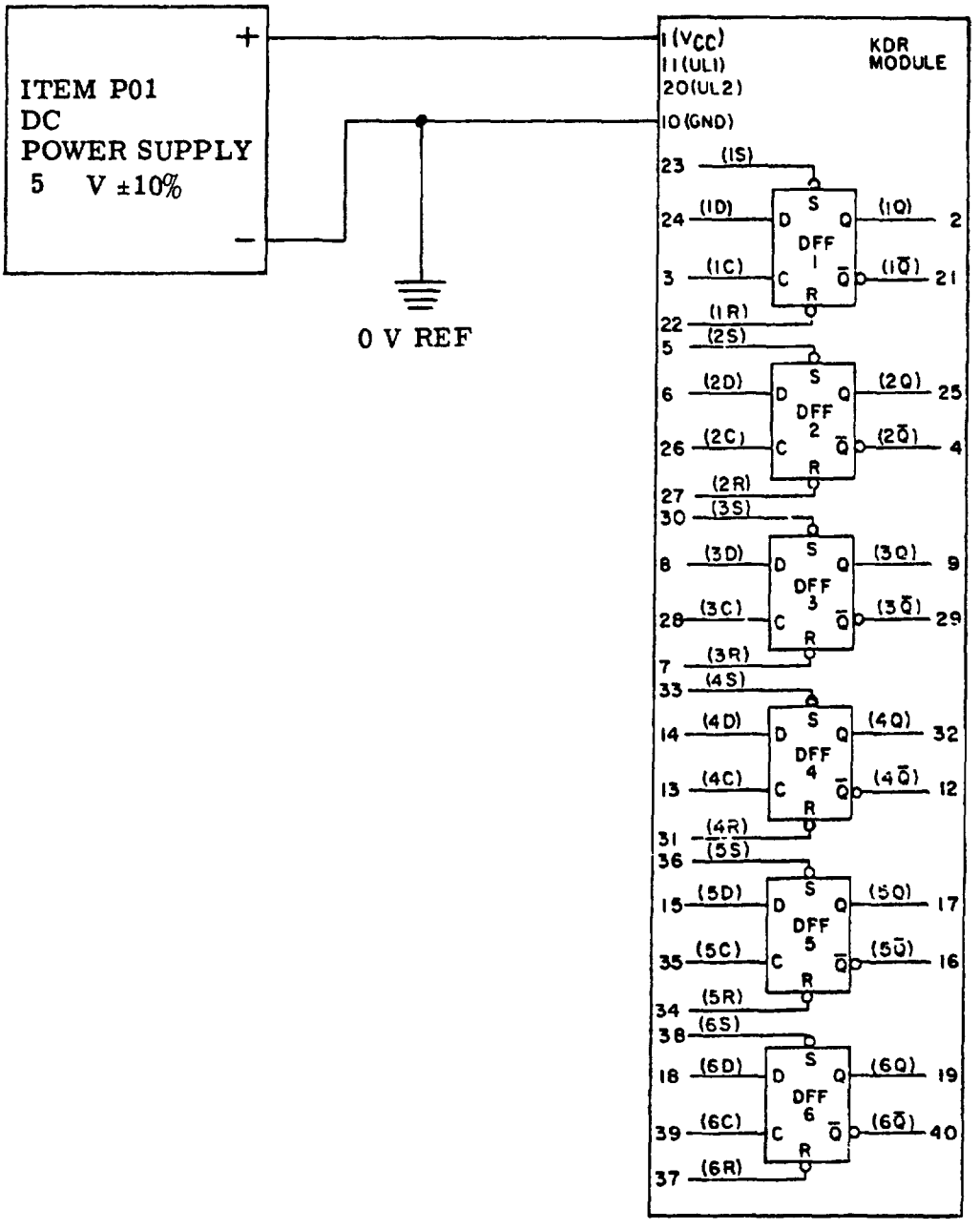


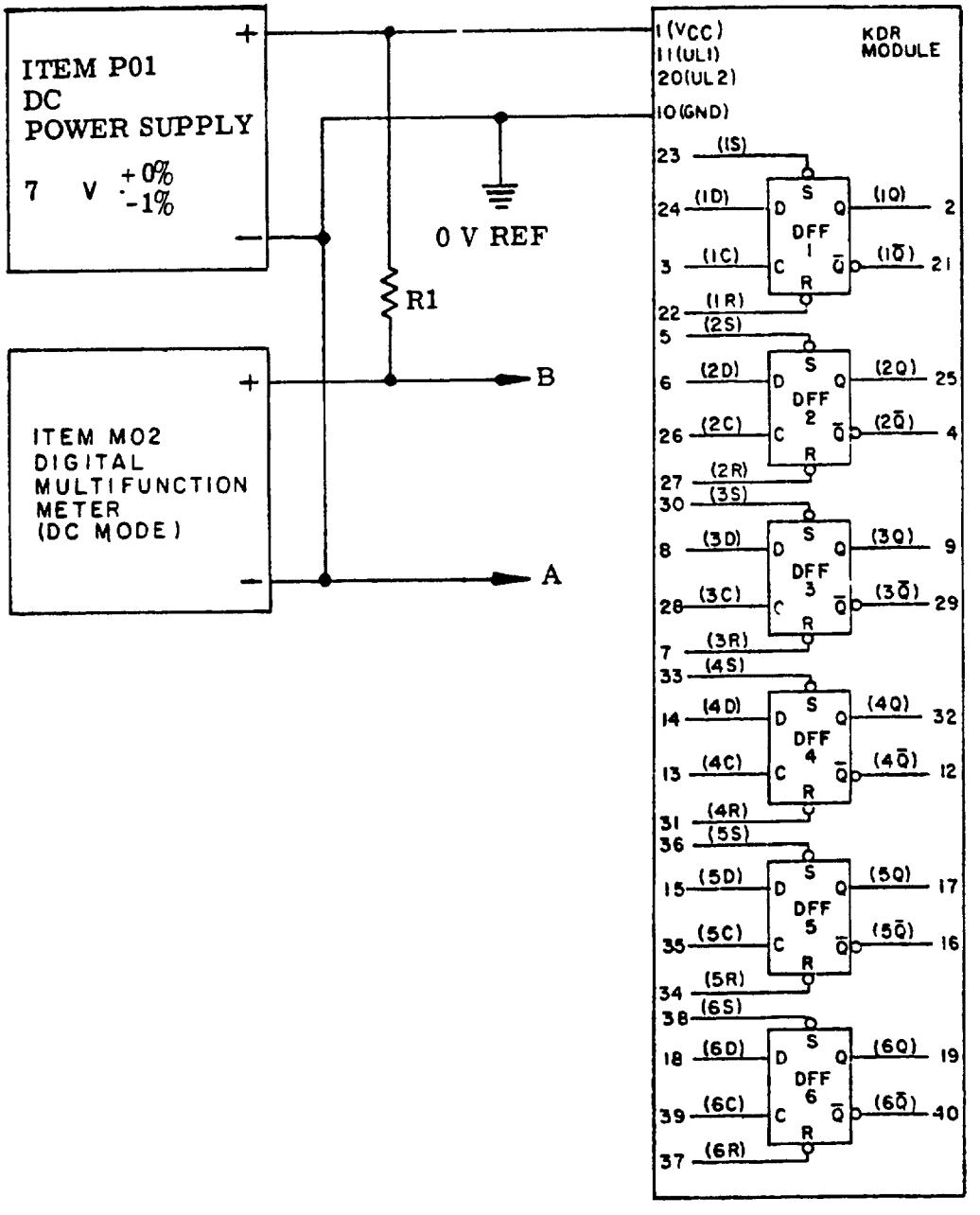
FIGURE 5. Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	1Q V_{OUT} (low loaded)	22	2	10	800	mV	6
A2	$1\overline{Q}$ "	23	21	"	"	"	"
A3	2Q "	27	25	"	"	"	"
A4	$2\overline{Q}$ "	5	4	"	"	"	"
A5	3Q "	7	9	"	"	"	"
A6	$3\overline{Q}$ "	30	29	"	"	"	"
A7	4Q "	31	32	"	"	"	"
A8	$4\overline{Q}$ "	33	12	"	"	"	"
A9	5Q "	34	17	"	"	"	"
A10	$5\overline{Q}$ "	36	16	"	"	"	"
A11	6Q "	37	19	"	"	"	"
A12	$6\overline{Q}$ "	38	40	"	"	"	"

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NOTES:
1. R1 = 316Ω ±1%, 1/4 W.
2. Measurements shall be recorded after point "B" has been connected to the appropriate module pin for at least 5 seconds.

* FIGURE 6. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input breakdown current test.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
B1	1D	Input breakdown current	24	22	Open	0	1	mA	7
B2	1C	"	3	22	"	"	"	"	"
B3	2D	"	6	27	"	"	"	"	"
B4	2C	"	26	27	"	"	"	"	"
B5	3D	"	8	7	"	"	"	"	"
B6	3C	"	28	7	"	"	"	"	"
B7	4D	"	14	31	"	"	"	"	"
B8	4C	"	13	31	"	"	"	"	"
B9	5D	"	15	34	"	"	"	"	"
B10	5C	"	35	34	"	"	"	"	"
B11	6D	"	18	37	"	"	"	"	"
B12	6C	"	39	37	"	"	"	"	"
B13	1S	"	23	Open	3	"	"	"	"
B14	2S	"	5	"	26	"	"	"	"
B15	3S	"	30	"	28	"	"	"	"
B16	4S	"	33	"	13	"	"	"	"
B17	5S	"	36	"	35	"	"	"	"
B18	6S	"	38	"	39	"	"	"	"

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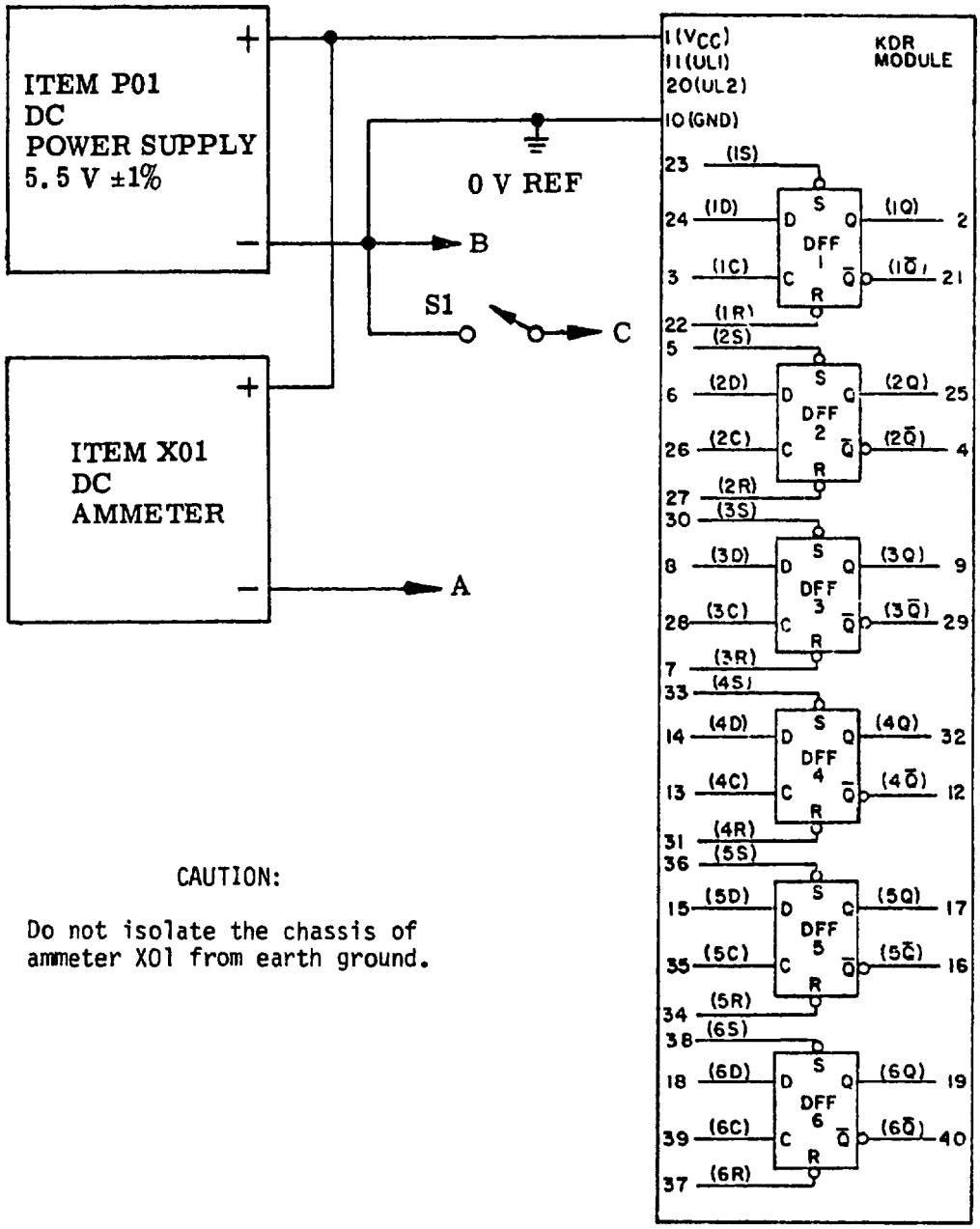
TABLE IV. Input breakdown current test - Continued.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
B19	1R	Input breakdown current	22	3,24	2	0.00	1.25	mA	7
B20	2R	"	27	6,26	25	"	"	"	"
B21	3R	"	7	8,28	9	"	"	"	"
B22	4R	"	31	13,14	32	"	"	"	"
B23	5R	"	34	15,35	17	"	"	"	"
B24	6R	"	37	18,39	19	"	"	"	"

TABLE V. Isolation test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
C	Isolation	Frame	10	∞	MΩ	8

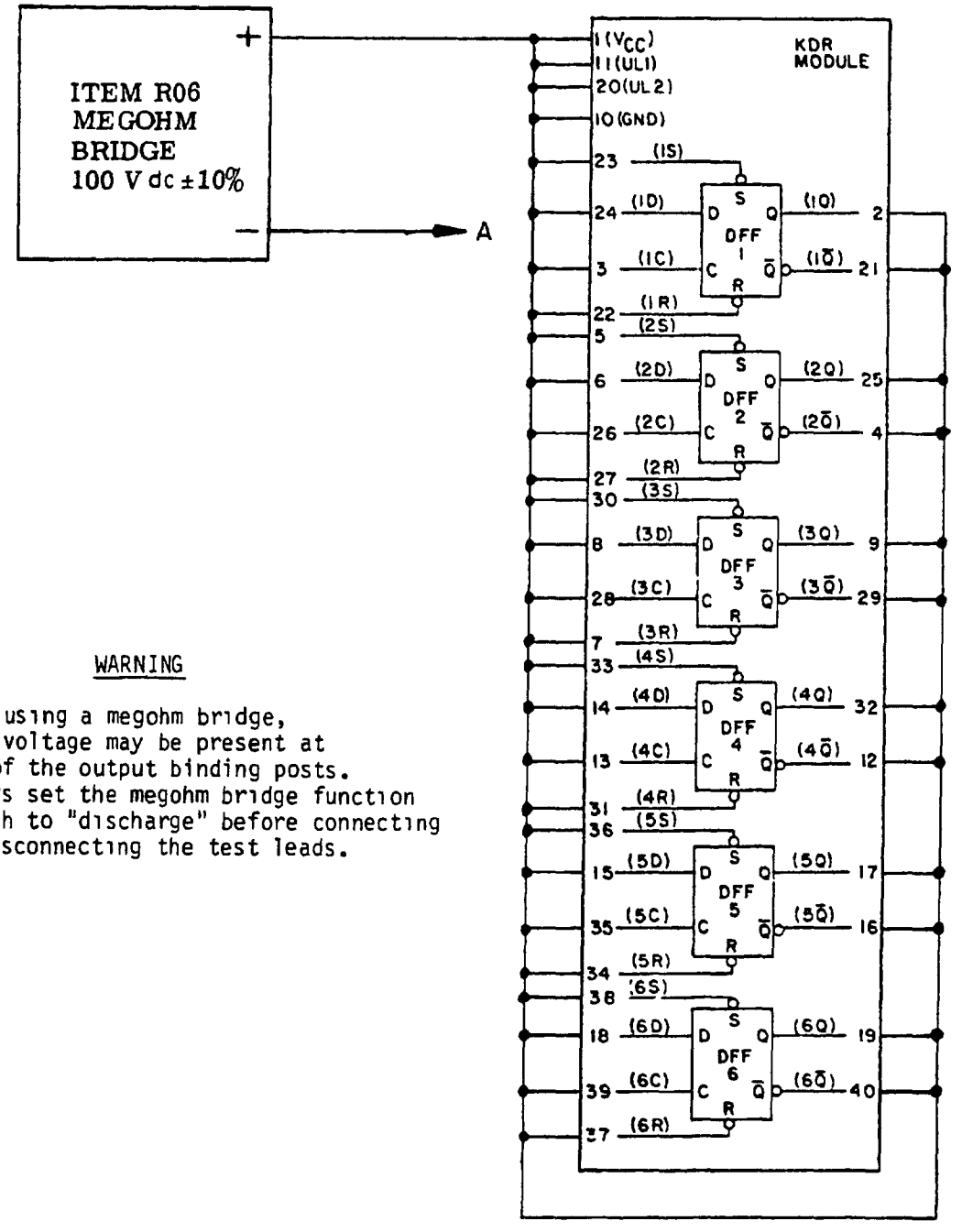
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NOTES:
1. S1 = SPST switch.
2. For electrical requirements B13-B18, set up for each test condition with S1 closed; after at least 100 ns, open S1.
3. For electrical requirements B19-B24, set up for each test condition with S1 open. Close S1 for at least 100 ns, then open S1.

FIGURE 7. Input breakdown current test circuit for table IV.

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 8. Isolation test circuit for table V.

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TABLE VI. Input capacitance test.

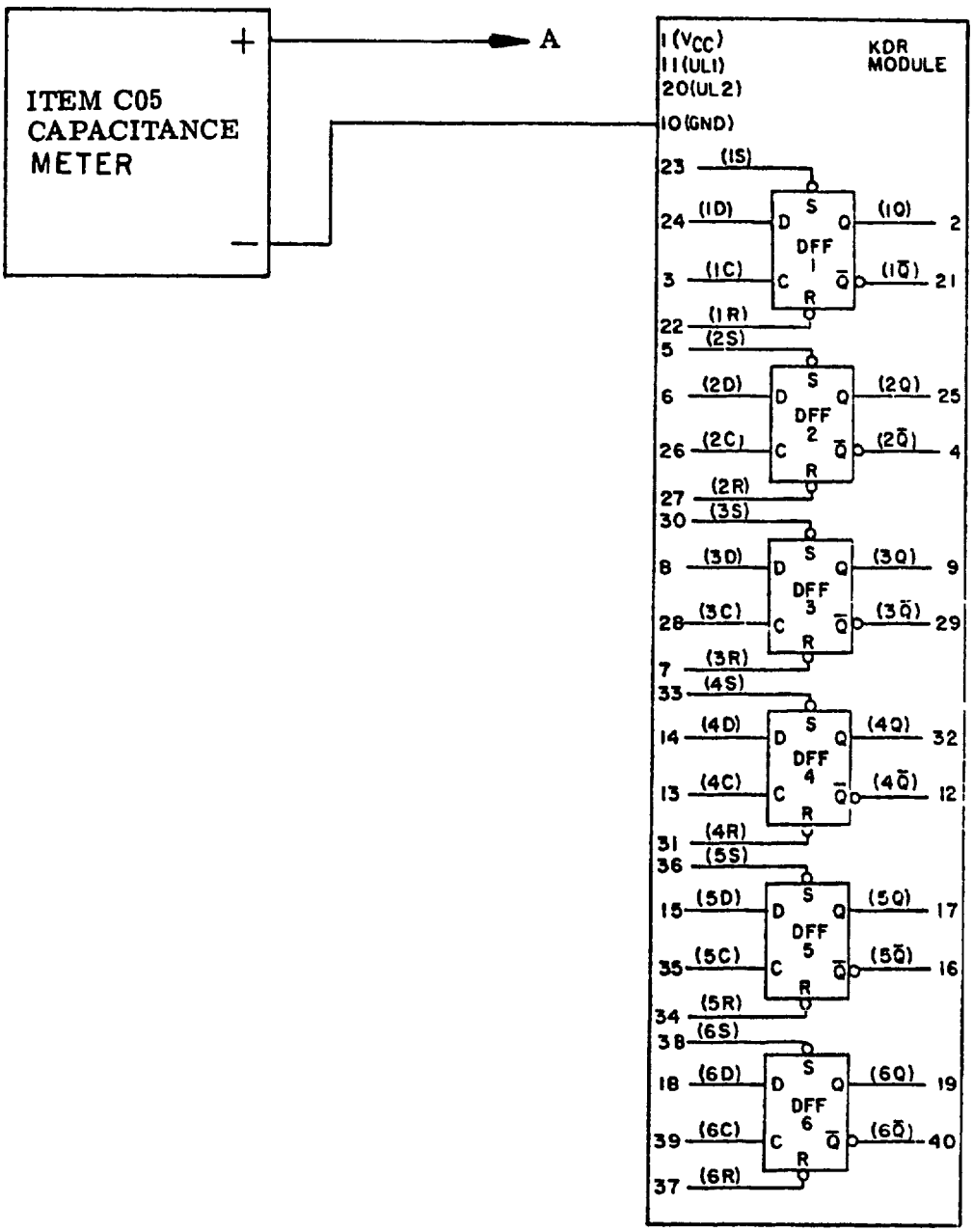
Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
D1	1C	Input capacitance	3	2	18	pF	9
D2	1S	"	23	"	"	"	"
D3	1R	"	22	"	"	"	"
D4	1D	"	24	"	"	"	"
D5	2S	"	5	"	"	"	"
D6	2D	"	6	"	"	"	"
D7	2R	"	27	"	"	"	"
D8	2C	"	26	"	"	"	"
D9	3D	"	8	"	"	"	"
D10	3C	"	28	"	"	"	"
D11	3S	"	30	"	"	"	"
D12	3R	"	7	"	"	"	"
D13	4C	"	13	"	"	"	"
D14	4D	"	14	"	"	"	"
D15	4S	"	33	"	"	"	"
D16	4R	"	31	"	"	"	"
D17	5D	"	15	"	"	"	"
D18	5C	"	35	"	"	"	"

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TABLE VI. Input capacitance test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
D19	5S	Input capacitance	36	2	18	pF	9
D20	5R	"	34	"	"	"	"
D21	6D	"	18	"	"	"	"
D22	6S	"	38	"	"	"	"
D23	6C	"	39	"	"	"	"
D24	6R	"	37	"	"	"	"

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

* FIGURE 9. Input capacitance test circuit for table VI.

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TABLE VII. Output voltage (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E1	1Q Output voltage, low level	23	3,22,24	2	21	0.0	0.4	V	10
F1	1$\overline{Q}$ Output voltage, high level	23	3,22,24	21	2	2.4	4.5	"	"
F2	1Q Output voltage, high level	Open	3,22,23,24	2	21	2.4	4.5	"	"
F3	1$\overline{Q}$ Output voltage, high level	Open	3,22,23,24	21	2	2.4	4.5	"	"
F4	1Q Output voltage, high level	22	3,23,24	2	21	2.4	4.5	"	"
E2	1$\overline{Q}$ Output voltage, low level	22	3,23,24	21	2	0.0	0.4	"	"
E3	1$\overline{Q}$ Output voltage, low level	3,22	23,24	21	2	0.0	0.4	"	"
F5	1Q Output voltage, high level	3,22	23,24	2	21	2.4	4.5	"	"
F6	1Q Output voltage, high level	24	3,22,23	2	21	2.4	4.5	"	"
F7	1$\overline{Q}$ Output voltage, high level	24	3,22,23	21	2	2.4	4.5	"	"
F8	1$\overline{Q}$ Output voltage, high level	23,24	3,22	21	2	2.4	4.5	"	"
E4	1Q Output voltage, low level	23,24	3,22	2	21	0.0	0.4	"	"
E5	1Q Output voltage, low level	3,23,24	22	2	21	0.0	0.4	"	"
F9	1$\overline{Q}$ Output voltage, high level	3,23,24	22	21	2	2.4	4.5	"	"
F10	1$\overline{Q}$ Output voltage, high level	22,23,24	3	21	2	2.4	4.5	"	"
E6	1Q Output voltage, low level	22,23,24	3	2	21	0.0	0.4	"	"
E7	1$\overline{Q}$ Output voltage, low level	3,22,23,24	Open	21	2	0.0	0.4	"	"

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TABLE VII. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F11	1Q Output voltage, high level	3,22,23,24	Open	2	21	2.4	4.5	V	10
F12	1Q Output voltage, high level	22,23	3,24	2	21	2.4	4.5	"	"
E8	1$\overline{Q}$ Output voltage, low level	22,23	3,24	21	2	0.0	0.4	"	"
E9	1Q Output voltage, low level	3,22,23	24	2	21	0.0	0.4	"	"
F13	1$\overline{Q}$ Output voltage, high level	3,22,23	24	21	2	2.4	4.5	"	"
E10	2Q Output voltage, low level	5	6,26,27	25	4	0.0	0.4	"	"
F14	2$\overline{Q}$ Output voltage, high level	5	6,26,27	4	25	2.4	4.5	"	"
F15	2Q Output voltage, high level	Open	5,6,26,27	25	4	2.4	4.5	"	"
F16	2$\overline{Q}$ Output voltage, high level	Open	5,6,26,27	4	25	2.4	4.5	"	"
F17	2Q Output voltage, high level	27	5,6,26	25	4	2.4	4.5	"	"
E11	2$\overline{Q}$ Output voltage, low level	27	5,6,26	4	25	0.0	0.4	"	"
E12	2$\overline{Q}$ Output voltage, low level	26,27	5,6	4	25	0.0	0.4	"	"
F18	2Q Output voltage, high level	26,27	5,6	25	4	2.4	4.5	"	"
F19	2Q Output voltage, high level	6	5,26,27	25	4	2.4	4.5	"	"
F20	2$\overline{Q}$ Output voltage, high level	6	5,26,27	4	25	2.4	4.5	"	"
F21	2$\overline{Q}$ Output voltage, high level	5,6	26,27	4	25	2.4	4.5	"	"
E13	2Q Output voltage, low level	5,6	26,27	25	4	0.0	0.4	"	"

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TABLE VII. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E14	2Q Output voltage, low level	5,6,26	27	25	4	0.0	0.4	V	10
F22	2$\overline{Q}$ Output voltage, high level	5,6,26	27	4	25	2.4	4.5	"	"
F23	2$\overline{Q}$ Output voltage, high level	5,6,27	26	4	25	2.4	4.5	"	"
E15	2Q Output voltage, low level	5,6,27	26	25	4	0.0	0.4	"	"
E16	2$\overline{Q}$ Output voltage, low level	5,6,26,27	Open	4	25	0.0	0.4	"	"
F24	2Q Output voltage, high level	5,6,26,27	Open	25	4	2.4	4.5	"	"
F25	2Q Output voltage, high level	5,27	6,26	25	4	2.4	4.5	"	"
E17	2$\overline{Q}$ Output voltage, low level	5,27	6,26	4	25	0.0	0.4	"	"
E18	2Q Output voltage, low level	5,26,27	6	25	4	0.0	0.4	"	"
F26	2$\overline{Q}$ Output voltage, high level	5,26,27	6	4	25	2.4	4.5	"	"
E19	3Q Output voltage, low level	30	8,7,28	9	29	0.0	0.4	"	"
F27	3$\overline{Q}$ Output voltage, high level	30	8,7,28	29	9	2.4	4.5	"	"
F28	3Q Output voltage, high level	Open	8,7,28,30	9	29	2.4	4.5	"	"
F29	3$\overline{Q}$ Output voltage, high level	Open	8,7,28,30	29	9	2.4	4.5	"	"
F30	3Q Output voltage, high level	7	8,28,30	9	29	2.4	4.5	"	"
E20	3$\overline{Q}$ Output voltage, low level	7	8,28,30	29	9	0.0	0.4	"	"
E21	3$\overline{Q}$ Output voltage, low level	7,28	8,30	29	9	0.0	0.4	"	"

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TABLE VII. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F31	3Q Output voltage, high level	7,28	8,30	9	29	2.4	4.5	V	10
F32	3Q Output voltage, high level	8	7,28,30	9	29	2.4	4.5	"	"
F33	3$\overline{Q}$ Output voltage, high level	8	7,28,30	29	9	2.4	4.5	"	"
F34	3$\overline{Q}$ Output voltage, high level	30,8	7,28	29	9	2.4	4.5	"	"
E22	3Q Output voltage, low level	30,8	7,28	9	29	0.0	0.4	"	"
E23	3Q Output voltage, low level	8,28,30	7	9	29	0.0	0.4	"	"
F35	3$\overline{Q}$ Output voltage, high level	8,28,30	7	29	9	2.4	4.5	"	"
F36	3$\overline{Q}$ Output voltage, high level	8,7,30	28	29	9	2.4	4.5	"	"
E24	3Q Output voltage, low level	8,7,30	28	9	29	0.0	0.4	"	"
E25	3$\overline{Q}$ Output voltage, low level	8,7,28,30	Open	29	9	0.0	0.4	"	"
F37	3Q Output voltage, high level	8,7,28,30	Open	9	29	2.4	4.5	"	"
F38	3Q Output voltage, high level	7,30	8,28	9	29	2.4	4.5	"	"
E26	3$\overline{Q}$ Output voltage, low level	7,30	8,28	29	9	0.0	0.4	"	"
E27	3Q Output voltage, low level	7,28,30	8	9	29	0.0	0.4	"	"
F39	3$\overline{Q}$ Output voltage, high level	7,28,30	8	29	9	2.4	4.5	"	"
E28	4Q Output voltage, low level	33	13,14,31	32	12	0.0	0.4	"	"
F40	4$\overline{Q}$ Output voltage, high level	33	13,14,31	12	32	2.4	4.5	"	"

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TABLE VII. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F41	4Q Output voltage, high level	Open	13,14,31,33	32	12	2.4	4.5	V	10
F42	4$\overline{Q}$ Output voltage, high level	Open	13,14,31,33	12	32	2.4	4.5	"	"
F43	4Q Output voltage, high level	31	13,14,33	32	12	2.4	4.5	"	"
E29	4$\overline{Q}$ Output voltage, low level	31	13,14,33	12	32	0.0	0.4	"	"
E30	4$\overline{Q}$ Output voltage, low level	13,31	14,33	12	32	0.0	0.4	"	"
F44	4Q Output voltage, high level	13,31	14,33	32	12	2.4	4.5	"	"
F45	4Q Output voltage, high level	14	13,31,33	32	12	2.4	4.5	"	"
F46	4$\overline{Q}$ Output voltage, high level	14	13,31,33	12	32	2.4	4.5	"	"
F47	4$\overline{Q}$ Output voltage, high level	14,33	13,31	12	32	2.4	4.5	"	"
E31	4Q Output voltage, low level	14,33	13,31	32	12	0.0	0.4	"	"
E32	4Q Output voltage, low level	13,14,33	31	32	12	0.0	0.4	"	"
F48	4$\overline{Q}$ Output voltage, high level	13,14,33	31	12	32	2.4	4.5	"	"
F49	4$\overline{Q}$ Output voltage, high level	14,31,33	13	12	32	2.4	4.5	"	"
E33	4Q Output voltage, low level	14,31,33	13	32	12	0.0	0.4	"	"
E34	4$\overline{Q}$ Output voltage, low level	13,14,31,33	Open	12	32	0.0	0.4	"	"
F50	4Q Output voltage, high level	13,14,31,33	Open	32	12	2.4	4.5	"	"
F51	4Q Output voltage, high level	31,33	13,14	32	12	2.4	4.5	"	"

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TABLE VII. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E35	4$\overline{Q}$ Output voltage, low level	31,33	13,14	12	32	0.0	0.4	V	10
E36	4Q Output voltage, low level	13,31,33	14	32	12	0.0	0.4	"	"
F52	4$\overline{Q}$ Output voltage, high level	13,31,33	14	12	32	2.4	4.5	"	"
E37	5Q Output voltage, low level	36	15,34,35	17	16	0.0	0.4	"	"
F53	5$\overline{Q}$ Output voltage, high level	36	15,34,35	16	17	2.4	4.5	"	"
F54	5Q Output voltage, high level	Open	15,34,35,36	17	16	2.4	4.5	"	"
F55	5$\overline{Q}$ Output voltage, high level	Open	15,34,35,36	16	17	2.4	4.5	"	"
F56	5Q Output voltage, high level	34	15,35,36	17	16	2.4	4.5	"	"
E38	5$\overline{Q}$ Output voltage, low level	34	15,35,36	16	17	0.0	0.4	"	"
E39	5$\overline{Q}$ Output voltage, low level	34,35	15,36	16	17	0.0	0.4	"	"
F57	5Q Output voltage, high level	34,35	15,36	17	16	2.4	4.5	"	"
F58	5Q Output voltage, high level	15	34,35,36	17	16	2.4	4.5	"	"
F59	5$\overline{Q}$ Output voltage, high level	15	34,35,36	16	17	2.4	4.5	"	"
F60	5$\overline{Q}$ Output voltage, high level	15,36	34,35	16	17	2.4	4.5	"	"
E40	5Q Output voltage, low level	15,36	34,35	17	16	0.0	0.4	"	"
E41	5Q Output voltage, low level	15,35,36	34	17	16	0.0	0.4	"	"
F61	5$\overline{Q}$ Output voltage, high level	15,35,36	34	16	17	2.4	4.5	"	"

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TABLE VII. Output voltage (low-level and high-level) test - Continued.

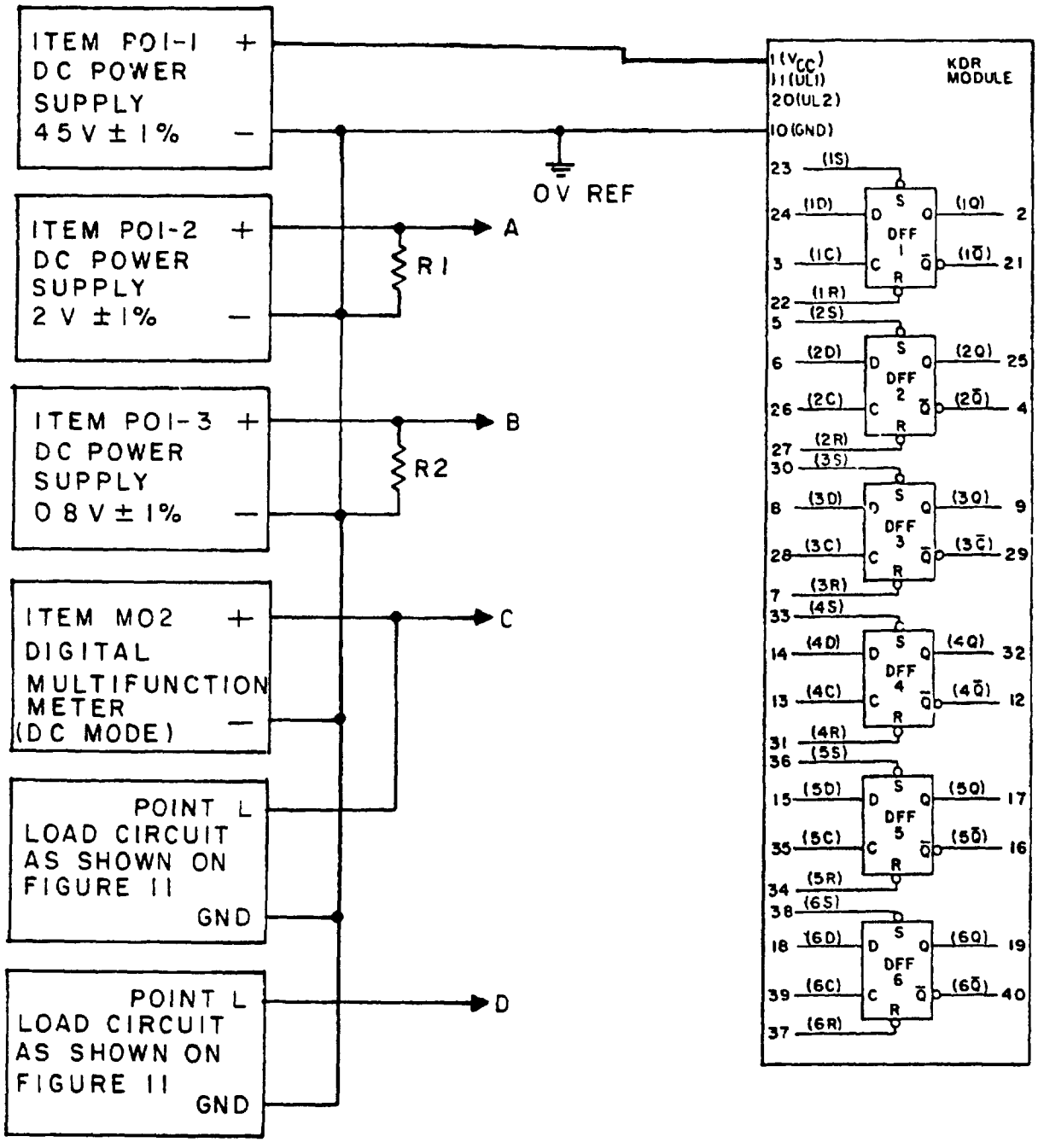
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F62	5$\overline{Q}$ Output voltage, high level	15,34,36	35	16	17	2.4	4.5	V	10
E42	5Q Output voltage, low level	15,34,36	35	17	16	0.0	0.4	"	"
E43	5$\overline{Q}$ Output voltage, low level	15,34,35,36	Open	16	17	0.0	0.4	"	"
F63	5Q Output voltage, high level	15,34,35,36	Open	17	16	2.4	4.5	"	"
F64	5Q Output voltage, high level	34,36	15,35	17	16	2.4	4.5	"	"
E44	5$\overline{Q}$ Output voltage, low level	34,36	15,35	16	17	0.0	0.4	"	"
E45	5Q Output voltage, low level	34,35,36	15	17	16	0.0	0.4	"	"
F65	5$\overline{Q}$ Output voltage, high level	34,35,36	15	16	17	2.4	4.5	"	"
E46	6Q Output voltage, low level	38	18,37,39	19	40	0.0	0.4	"	"
F66	6$\overline{Q}$ Output voltage, high level	38	18,37,39	40	19	2.4	4.5	"	"
F67	6Q Output voltage, high level	Open	18,37,38,39	19	40	2.4	4.5	"	"
F68	6$\overline{Q}$ Output voltage, high level	Open	18,37,38,39	40	19	2.4	4.5	"	"
F69	6Q Output voltage, high level	37	18,38,39	19	40	2.4	4.5	"	"
E47	6$\overline{Q}$ Output voltage, low level	37	18,38,39	40	19	0.0	0.4	"	"
E48	6$\overline{Q}$ Output voltage, low level	37,39	18,38	40	19	0.0	0.4	"	"
F70	6Q Output voltage, high level	37,39	18,38	19	40	2.4	4.5	"	"
F71	6Q Output voltage, high level	18	37,38,39	19	40	2.4	4.5	"	"

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TABLE VII. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F72	$6\overline{Q}$ Output voltage, high level	18	37,38,39	40	19	2.4	4.5	V	10
F73	$6\overline{Q}$ Output voltage, high level	18,38	37,39	40	19	2.4	4.5	"	"
E49	6Q Output voltage, low level	18,38	37,39	19	40	0.0	0.4	"	"
E50	6Q Output voltage, low level	18,38,39	37	19	40	0.0	0.4	"	"
F74	$6\overline{Q}$ Output voltage, high level	18,38,39	37	40	19	2.4	4.5	"	"
F75	$6\overline{Q}$ Output voltage, high level	18,37,38	39	40	19	2.4	4.5	"	"
E51	6Q Output voltage, low level	18,37,38	39	19	40	0.0	0.4	"	"
E52	$6\overline{Q}$ Output voltage, low level	18,37,38,39	Open	40	19	0.0	0.4	"	"
F76	6Q Output voltage, high level	18,37,38,39	Open	19	40	2.4	4.5	"	"
F77	6Q Output voltage, high level	37,38	18,39	19	40	2.4	4.5	"	"
E53	$6\overline{Q}$ Output voltage, low level	37,38	18,39	40	19	0.0	0.4	"	"
E54	6Q Output voltage, low level	37,38,39	18	19	40	0.0	0.4	"	"
F78	$6\overline{Q}$ Output voltage, high level	37,38,39	18	40	19	2.4	4.5	"	"

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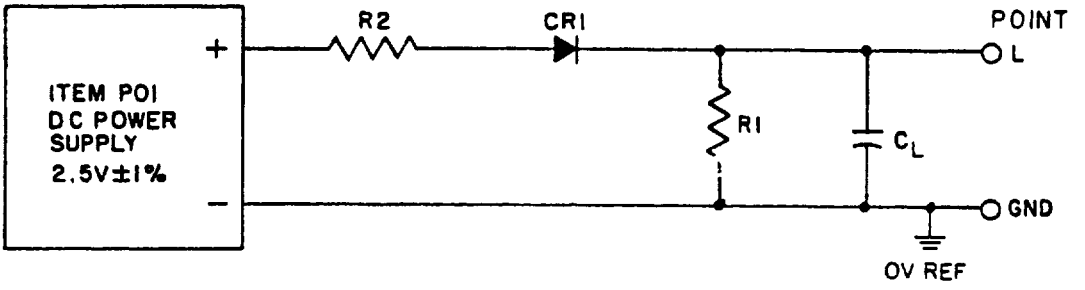


NOTES:

1. R1 = R2 = 51Ω ±5%, 1 W.
2. The electrical requirements for each circuit must be performed in the order listed in table VII since each electrical requirement depends on the previous test.
3. Both outputs of the circuit under test must be loaded simultaneously.
4. When the clock input (pins 3, 26, 28, 13, 35, and 39) is changed from high level to low level, multiple transitions (e.g. due to switch bounce) may result in a false output. In this event, a momentary low level may be applied to the "set" or "reset" input to correct the condition. When the clock input is changed from low level to high level, the proper state must occur without the use of a mor. entary "set" or "reset".

* FIGURE 10. Output voltage (low-level and high-level) test circuit for table VII.

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NOTES:
1. R1 = 6.04 kΩ ±1%, 1/8 W.
2. R2 = select value for current flow of 16 mA ±1% with point L held at 0.4 V.
3. CR1 = G321 (small case) or 1N4150 or equivalent.
4. C_L = 100 ±5 pF (including probe and parasitic capacitance) for test conditions L, M, N, P, T, U, V, and W.
5. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test condition X.
6. C_L is not required for dc electrical test conditions E and F.
7. The maximum lead length from point L to the appropriate module pin shall be 2 inches (50.80 mm).
8. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

FIGURE 11. Logic load circuit.

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TABLE VIII. Input current (low-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	1D Input current, low level	22	Open	3	24	0.40	1.73	mA	12
G2	2D "	27	"	26	6	"	"	"	"
G3	3D "	7	"	28	8	"	"	"	"
G4	4D "	31	"	13	14	"	"	"	"
G5	5D "	34	"	35	15	"	"	"	"
G6	6D "	37	"	39	18	"	"	"	"
G7	1S "	Open	"	22,24	23	0.80	3.46	"	"
G8	2S "	"	"	6,27	5	"	"	"	"
G9	3S "	"	"	7,8	30	"	"	"	"
G10	4S "	"	"	14,31	33	"	"	"	"
G11	5S "	"	"	15,34	36	"	"	"	"
G12	6S "	"	"	18,37	38	"	"	"	"
G13	1C "	22	"	23,24	3	"	"	"	"
G14	2C "	27	"	5,6	26	"	"	"	"
G15	3C "	7	"	8,30	28	"	"	"	"
G16	4C "	31	"	14,33	13	"	"	"	"
G17	5C "	34	"	15,36	35	"	"	"	"
G18	6C "	37	"	18,38	39	"	"	"	"

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TABLE VIII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G19	1R Input current, low level	24	3	23	22	1.20	5.19	mA	12
G20	2R "	6	26	5	27	"	"	"	"
G21	3R "	8	28	30	7	"	"	"	"
G22	4R "	14	13	33	31	"	"	"	"
G23	5R "	15	35	36	34	"	"	"	"
G24	6R "	18	39	38	37	"	"	"	"

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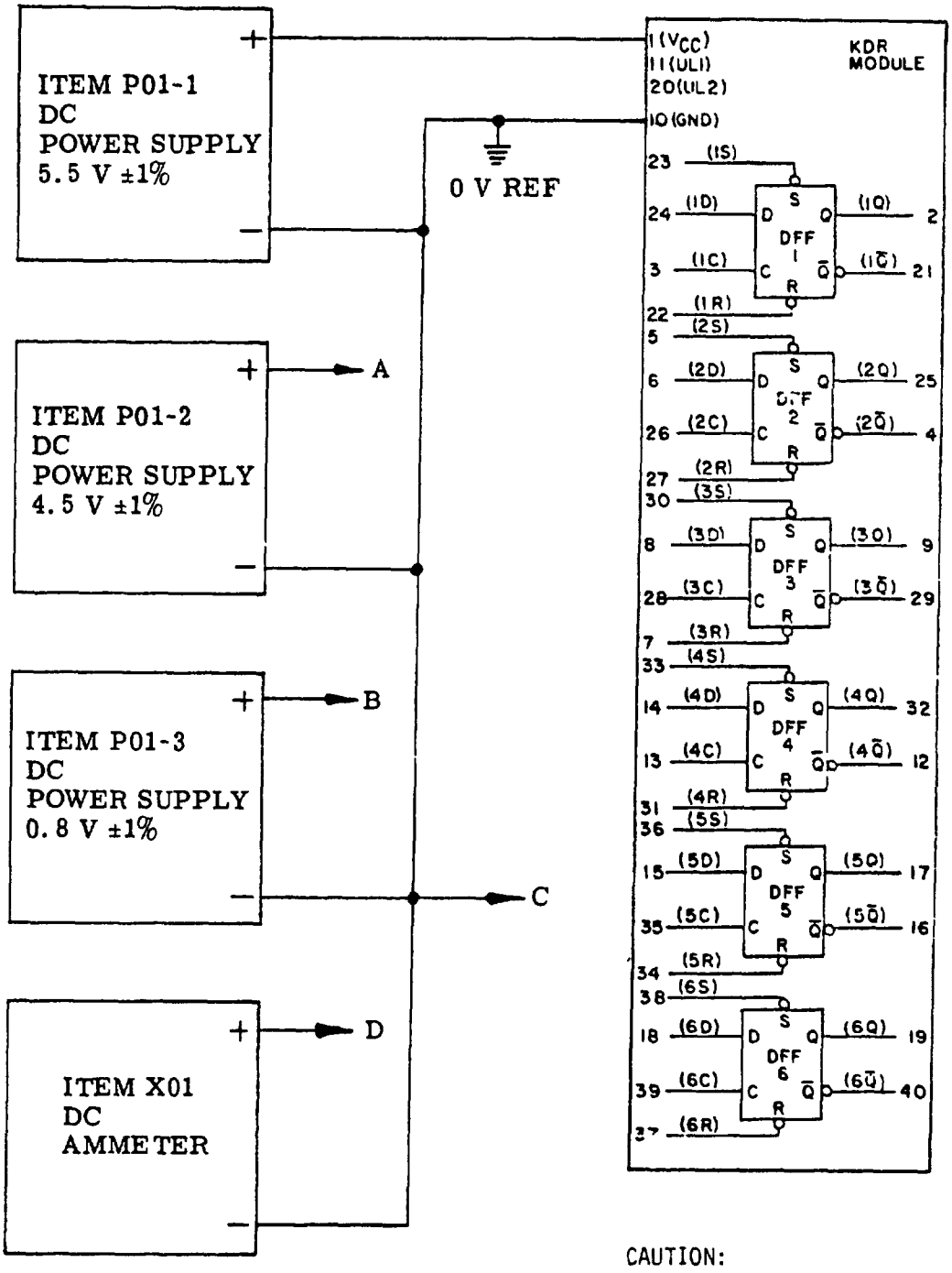


FIGURE 12. Input current (low-level) test circuit for table VIII.

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TABLE IX. Input current (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	1D Input current, high level	24	22	Open	Open	0	30	µA	13
H2	2D "	6	27	"	"	"	"	"	"
H3	3D "	8	7	"	"	"	"	"	"
H4	4D "	14	31	"	"	"	"	"	"
H5	5D "	15	34	"	"	"	"	"	"
H6	6D "	18	37	"	"	"	"	"	"
H7	1C "	3	22	"	"	"	60	"	"
H8	2C "	26	27	"	"	"	"	"	"
H9	3C "	28	7	"	"	"	"	"	"
H10	4C "	13	31	"	"	"	"	"	"
H11	5C "	35	34	"	"	"	"	"	"
H12	6C "	39	37	"	"	"	"	"	"
H13	1S "	23	Open	"	3	"	"	"	"
H14	2S "	5	"	"	26	"	"	"	"
H15	3S "	30	"	"	28	"	"	"	"
H16	4S "	33	"	"	13	"	"	"	"
H17	5S "	36	"	"	35	"	"	"	"
H18	6S "	38	"	"	39	"	"	"	"

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TABLE IX. Input current (high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H19	1R Input current, high level	22	3,24	Open	2	0	90	µA	13
H20	2R "	27	6,26	"	25	"	"	"	"
H21	3R "	7	8,28	"	9	"	"	"	"
H22	4R "	31	13,14	"	32	"	"	"	"
H23	5R "	34	15,35	"	17	"	"	"	"
H24	6R "	37	18,39	"	19	"	"	"	"

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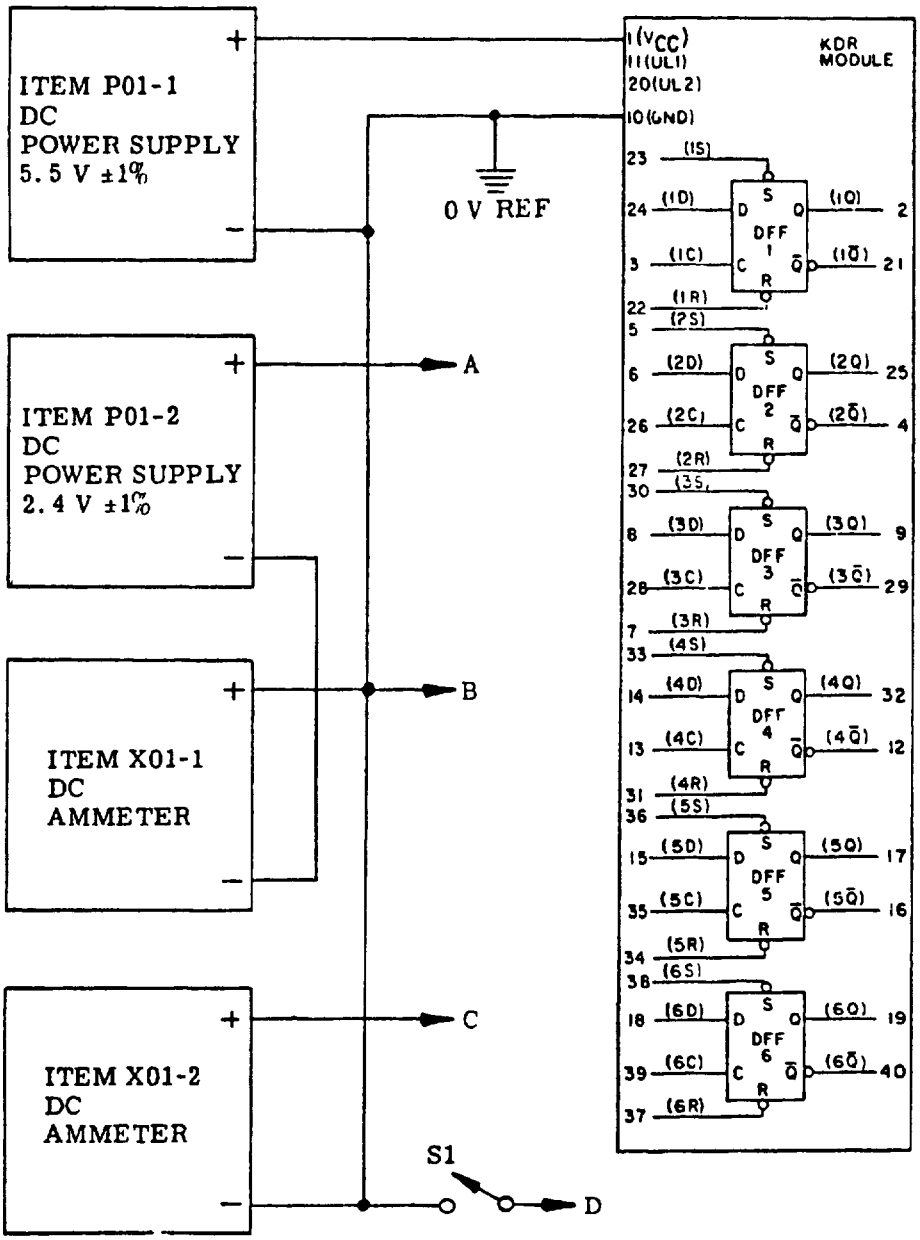
TABLE X. Output short circuit current test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
J1	1Q Output short circuit current	Open	23	2	Open	20	55	mA	13
J2	1$\overline{Q}$ "	"	22	21	"	"	"	"	"
J3	2Q "	"	5	25	"	"	"	"	"
J4	2$\overline{Q}$ "	"	27	4	"	"	"	"	"
J5	3Q "	"	30	9	"	"	"	"	"
J6	3$\overline{Q}$ "	"	7	29	"	"	"	"	"
J7	4Q "	"	33	32	"	"	"	"	"
J8	4$\overline{Q}$ "	"	31	12	"	"	"	"	"
J9	5Q "	"	36	17	"	"	"	"	"
J10	5$\overline{Q}$ "	"	34	16	"	"	"	"	"
J11	6Q "	"	38	19	"	"	"	"	"
J12	6$\overline{Q}$ "	"	37	40	"	"	"	"	"
J13	UL1 "	"	Open	11	"	5.1	5.9	"	"
J14	UL2 "	"	Open	20	"	5.1	5.9	"	"

TABLE XI. Power supply current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	Power supply current, low level	Open	Open	1	90	mA	14
K2	Power supply current, high level	B	A	1	90	mA	14

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CAUTION: Do not isolate the chassis of ammeter X01 from earth ground. If tests are successively repeated, a time of 10 seconds minimum between tests shall be allowed to preclude detrimental effects due to module heating.

NOTES:
1. S1 = SPST switch.
2. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required. Use item X01-1 for table X, use item X01-2 for table IX.
3. With point A disconnected, item X01-1 should indicate zero current on 1 µA range.
4. When performing output short circuit tests, only one output shall be shorted at a time and the measurement taken within 5 seconds.
5. For electrical requirements H13-H18, set up for each test condition with S1 closed; after 100 ns, open S1.
6. For electrical requirements H19-H24, set up for each test condition with S1 open. Close S1 for at least 100 ns, then open S1.

* Figure 13. Input current (high-level) and output short circuit current test circuit for tables IX and X.

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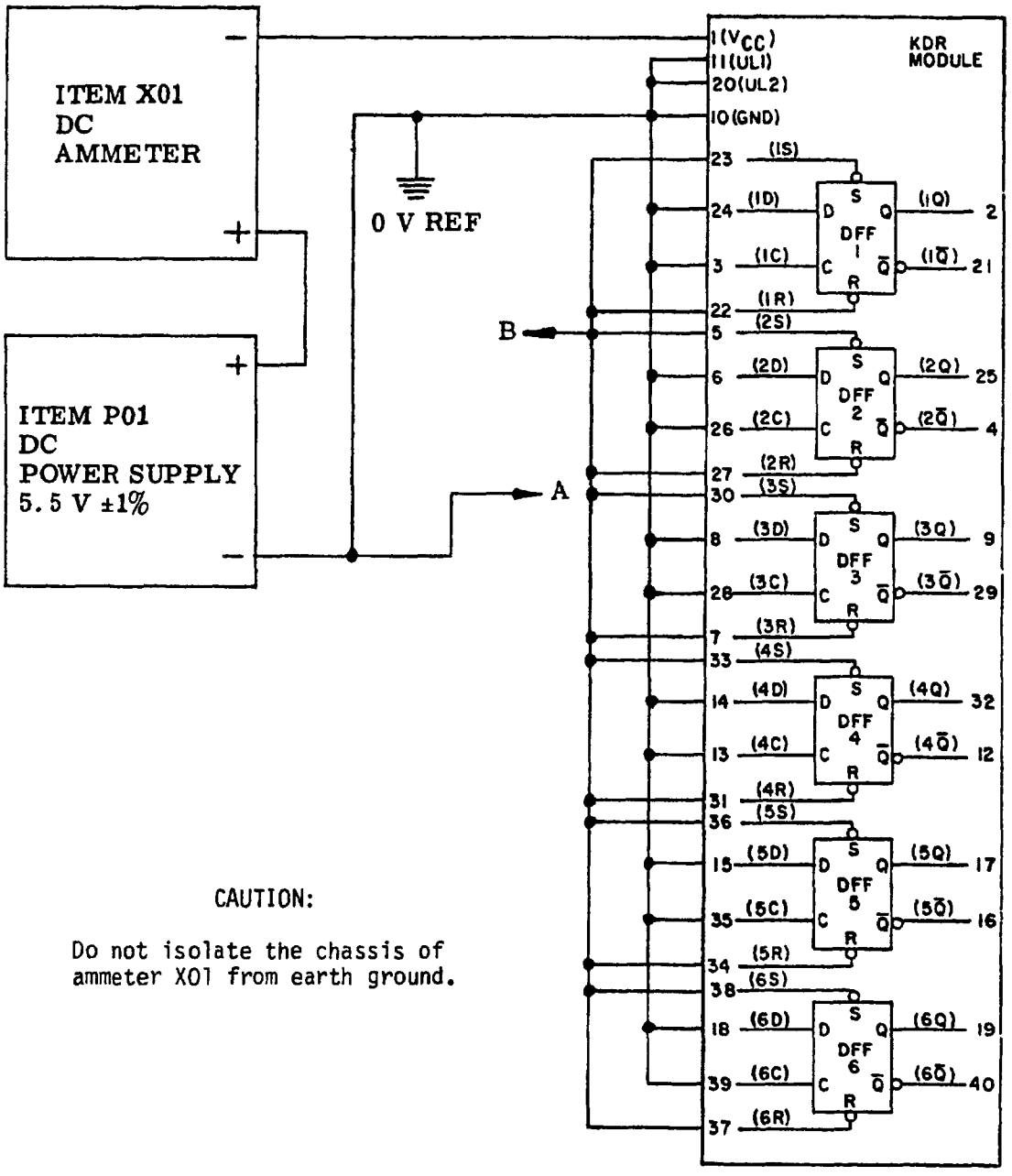


FIGURE 14. Power supply current test circuit for table XI.

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TABLE XII. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L1	1Q Transition time (t_{TLH})	3	24	2	21	4	45	ns	15,16
L2	1$\overline{Q}$ "	3	24	21	2	"	"	"	"
L3	2Q "	26	6	25	4	"	"	"	"
L4	2$\overline{Q}$ "	26	6	4	25	"	"	"	"
L5	3Q "	28	8	9	29	"	"	"	"
L6	3$\overline{Q}$ "	28	8	29	9	"	"	"	"
L7	4Q "	13	14	32	12	"	"	"	"
L8	4$\overline{Q}$ "	13	14	12	32	"	"	"	"
L9	5Q "	35	15	17	16	"	"	"	"
L10	5$\overline{Q}$ "	35	15	16	17	"	"	"	"
L11	6Q "	39	18	19	40	"	"	"	"
L12	6$\overline{Q}$ "	39	18	40	19	"	"	"	"
M1	1Q Transition time (t_{THL})	3	24	2	21	"	20	"	"
M2	1$\overline{Q}$ "	3	24	21	2	"	"	"	"
M3	2Q "	26	6	25	4	"	"	"	"
M4	2$\overline{Q}$ "	26	6	4	25	"	"	"	"
M5	3Q "	28	8	9	29	"	"	"	"
M6	3$\overline{Q}$ "	28	8	29	9	"	"	"	"
M7	4Q "	13	14	32	12	"	"	"	"
M8	4$\overline{Q}$ "	13	14	12	32	"	"	"	"
M9	5Q "	35	15	17	16	"	"	"	"

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TABLE XII. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M10	$5\overline{Q}$ "	35	15	16	17	4	20	ns	15,16
M11	6Q "	39	18	19	40	"	"	"	"
M12	$6\overline{Q}$ "	39	18	40	19	"	"	"	"

TABLE XIII. Propagation delay time (t_{PHL} and t_{PLH}) test.

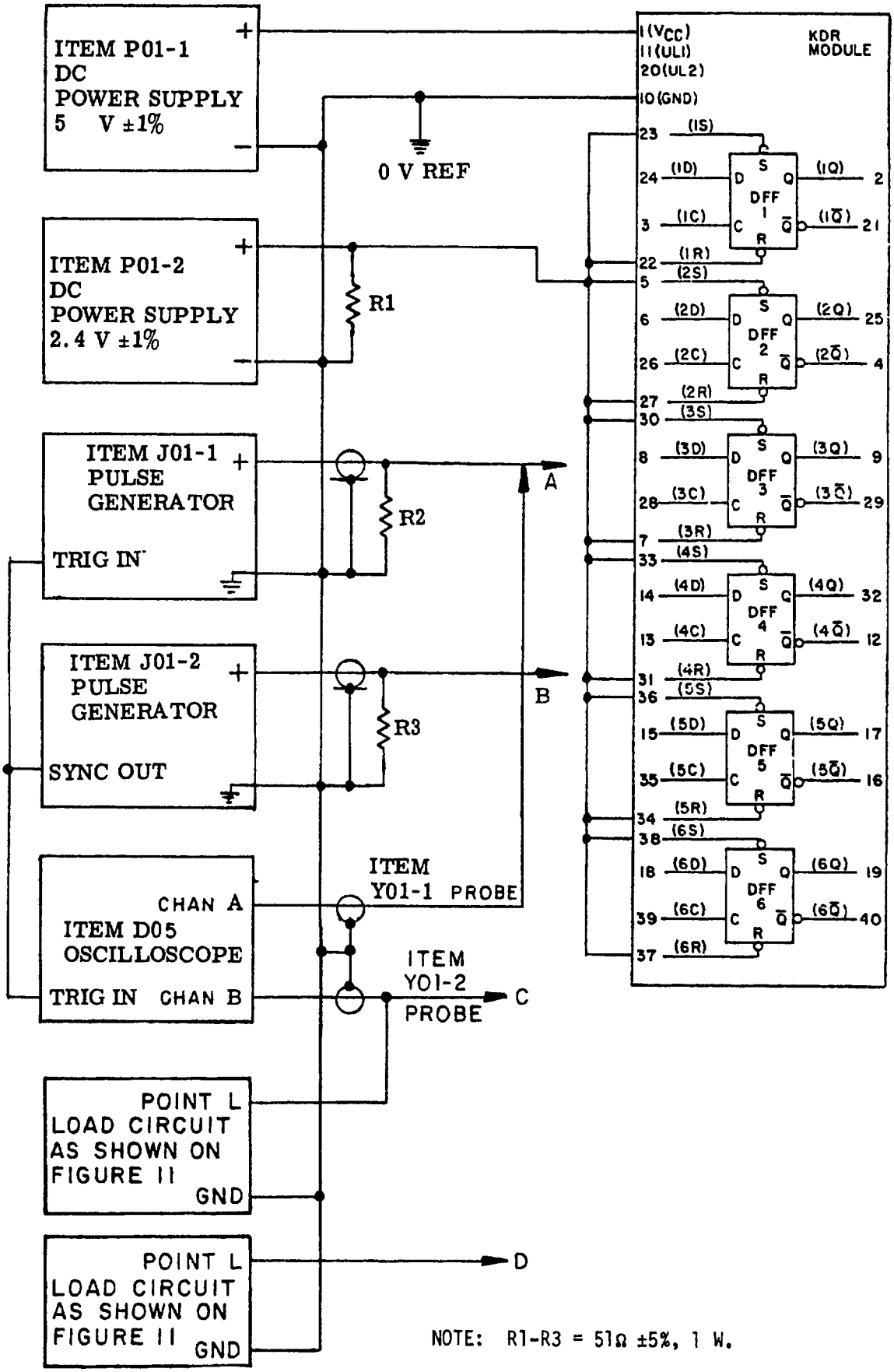
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	1Q Propagation delay time (t_{PHL})	3	24	2	21	6	50	ns	15,16
N2	$1\overline{Q}$ "	3	24	21	2	"	"	"	"
N3	2Q "	26	6	25	4	"	"	"	"
N4	$2\overline{Q}$ "	26	6	4	25	"	"	"	"
N5	3Q "	28	8	9	29	"	"	"	"
N6	$3\overline{Q}$ "	28	8	29	9	"	"	"	"
N7	4Q "	13	14	32	12	"	"	"	"
N8	$4\overline{Q}$ "	13	14	12	32	"	"	"	"
N9	5Q "	35	15	17	16	"	"	"	"
N10	$5\overline{Q}$ "	35	15	16	17	"	"	"	"
N11	6Q "	39	18	19	40	"	"	"	"
N12	$6\overline{Q}$ "	39	18	40	19	"	"	"	"

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TABLE XIII. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1Q Propagation delay time (t_{PLH})	3	24	2	21	6	30	ns	15,16
P2	$1\overline{Q}$ "	3	24	21	2	"	"	"	"
P3	2Q "	26	6	25	4	"	"	"	"
P4	$2\overline{Q}$ "	26	6	4	25	"	"	"	"
P5	3Q "	28	8	9	29	"	"	"	"
P6	$3\overline{Q}$ "	28	8	29	9	"	"	"	"
P7	4Q "	13	14	32	12	"	"	"	"
P8	$4\overline{Q}$ "	13	14	12	32	"	"	"	"
P9	5Q "	35	15	17	16	"	"	"	"
P10	$5\overline{Q}$ "	35	15	16	17	"	"	"	"
P11	6Q "	39	18	19	40	"	"	"	"
P12	$6\overline{Q}$ "	39	18	40	19	"	"	"	"

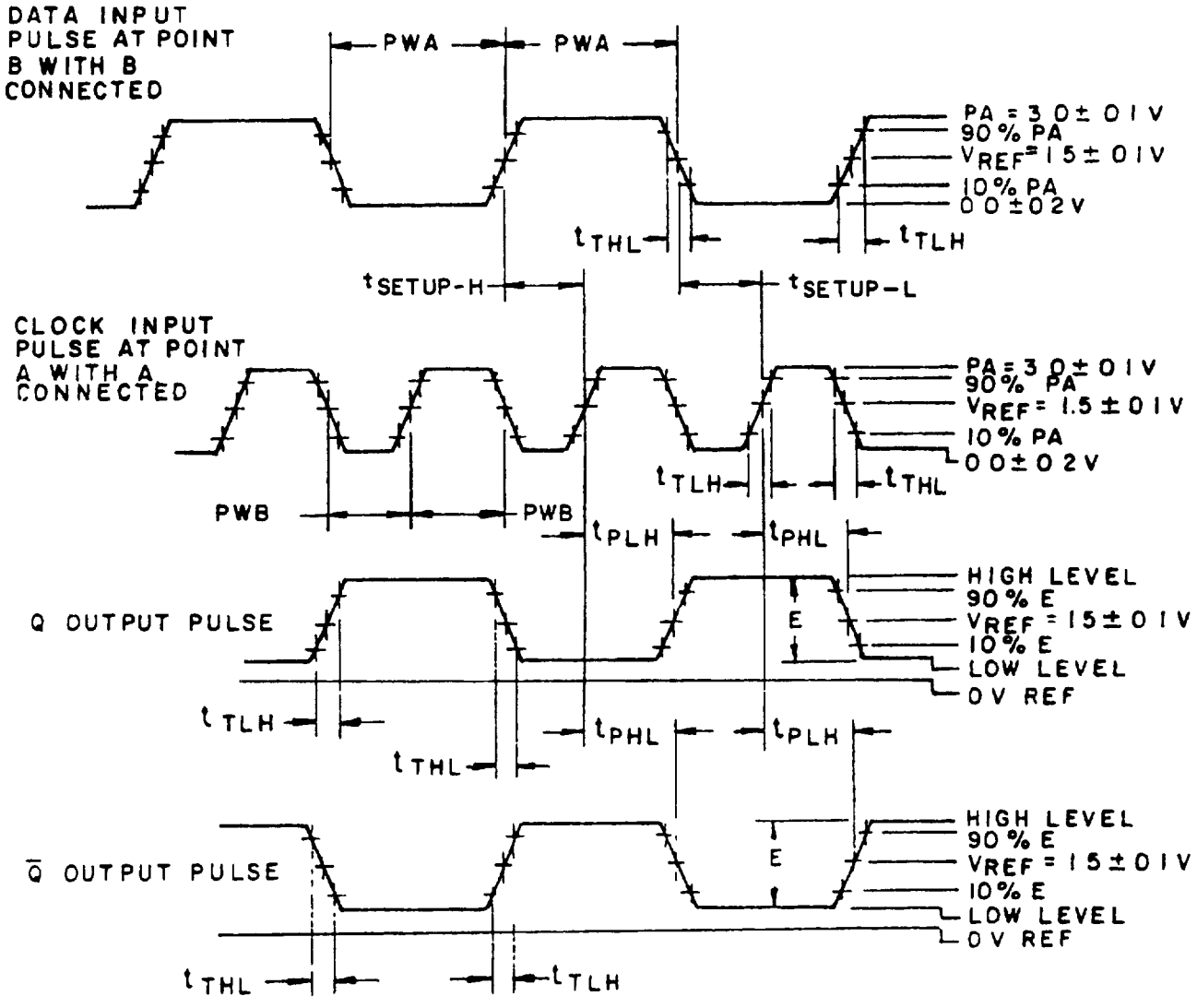
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NOTE: R1-R3 = 51Ω ±5%, 1 W.

* FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit for tables XII and XIII.

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Notes;
1. Input pulse parameters:
   PWA = 500 ns ±10%.
   PWB = 150 ns ±10%.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   $t_{SETUP-H} = t_{SETUP-L} = 125$ ns ±10%.
2. Both outputs of the circuit under test shall be loaded simultaneously.
3. Before making these measurements, the output levels shall be verified to be $\leq$ 0.4 V for the low level and $\geq$ 2.4 V for the high level.

* FIGURE 16. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables XII and XIII.

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TABLE XIV. Setup time and hold time test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient						Unit	Test figure
			A to pin	B to pin	C to pin	TA		TB		TC			
						Min	Max	Min	Max	Min	Max		
R1	1Q	Setup time	3	24	2	21	73	431	591	900	1100	ns	17,18
R2	2Q	"	26	6	25	"	"	"	"	"	"	"	"
R3	3Q	"	28	8	9	"	"	"	"	"	"	"	"
R4	4Q	"	13	14	32	"	"	"	"	"	"	"	"
R5	5Q	"	35	15	17	"	"	"	"	"	"	"	"
R6	6Q	"	39	18	19	"	"	"	"	"	"	"	"
R7	$1\overline{Q}$	"	3	24	21	471	623	"	"	"	"	"	"
R8	$2\overline{Q}$	"	26	6	4	"	"	"	"	"	"	"	"
R9	$3\overline{Q}$	"	28	8	29	"	"	"	"	"	"	"	"
R10	$4\overline{Q}$	"	13	14	12	"	"	"	"	"	"	"	"
R11	$5\overline{Q}$	"	35	15	16	"	"	"	"	"	"	"	"
R12	$6\overline{Q}$	"	39	18	40	"	"	"	"	"	"	"	"
S1	1Q	Hold time	3	24	2	450	598	"	"	"	"	"	17,19
S2	2Q	"	26	6	25	"	"	"	"	"	"	"	"
S3	3Q	"	28	8	9	"	"	"	"	"	"	"	"
S4	4Q	"	13	14	32	"	"	"	"	"	"	"	"
S5	5Q	"	35	15	17	"	"	"	"	"	"	"	"
S6	6Q	"	39	18	19	"	"	"	"	"	"	"	"
S7	$1\overline{Q}$	"	3	24	21	0	48	"	"	"	"	"	"
S8	$2\overline{Q}$	"	26	6	4	0	48	"	"	"	"	"	"

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TABLE XIV. Setup time and hold time test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient						Unit	Test figure
			A to pin	B to pin	C to pin	TA		TB		TC			
						Min	Max	Min	Max	Min	Max		
S9	$3\overline{Q}$	Hold time	28	8	29	0	48	431	591	900	1100	ns	17,19
S10	$4\overline{Q}$	"	13	14	12	"	"	"	"	"	"	"	"
S11	$5\overline{Q}$	"	35	15	16	"	"	"	"	"	"	"	"
S12	$6\overline{Q}$	"	39	18	40	"	"	"	"	"	"	"	"

TABLE XV. Maximum clock frequency/minimum clock pulse width test.

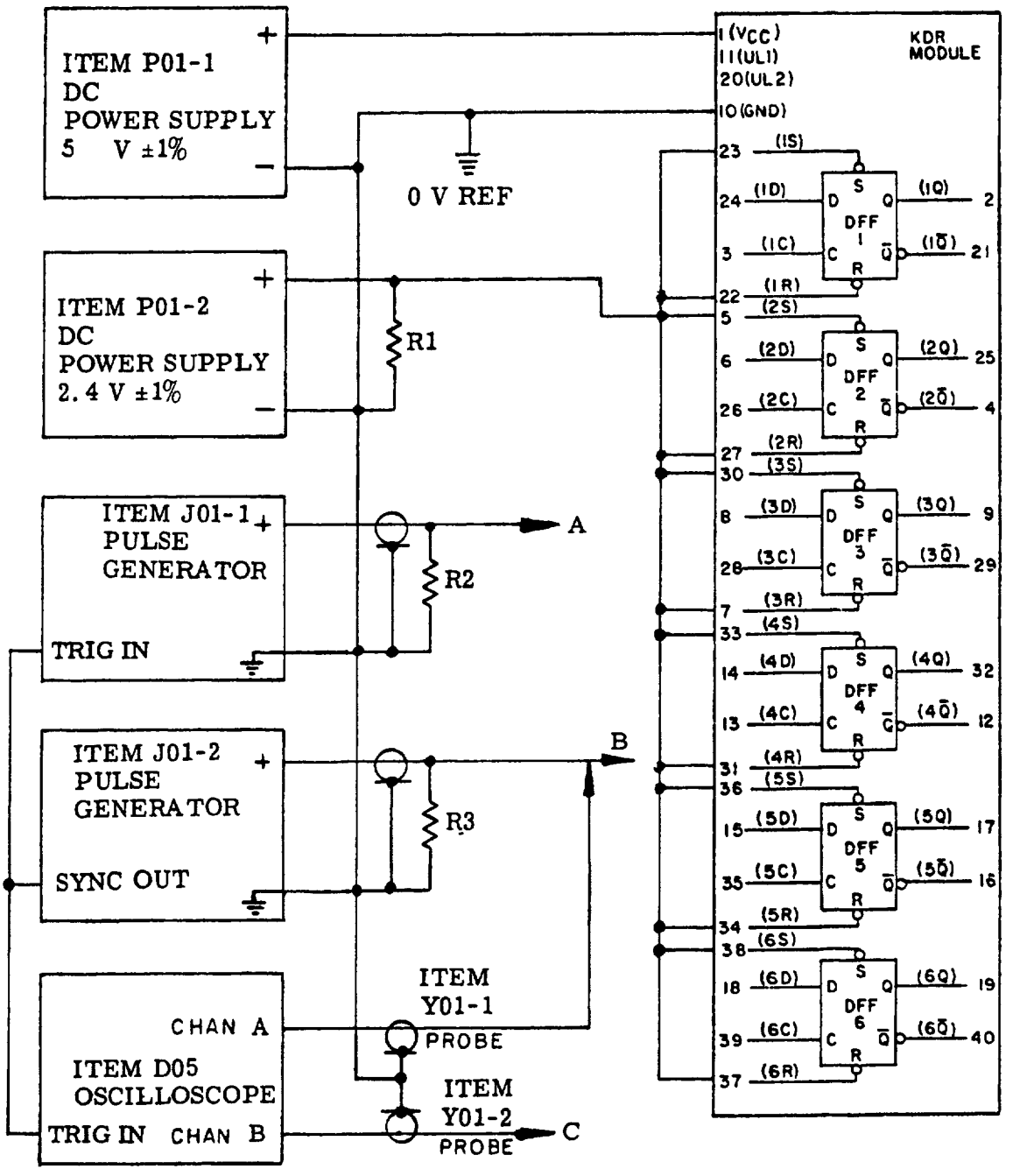
Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient						Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	TA		TB		TC			
							Min	Max	Min	Max	Min	Max		
T1	1Q	Maximum clock frequency/ *minimum clock pulse width*	3	24	2	21	60	115	73	121	160	176	ns	20,21
T2	2Q	"	26	6	25	4	"	"	"	"	"	"	"	"
T3	3Q	"	28	8	9	29	"	"	"	"	"	"	"	"
T4	4Q	"	13	14	32	12	"	"	"	"	"	"	"	"
T5	5Q	"	35	15	17	16	"	"	"	"	"	"	"	"
T6	6Q	"	39	18	19	40	"	"	"	"	"	"	"	"
T7	$1\overline{Q}$	"	3	24	21	2	140	203	"	"	"	"	"	"
T8	$2\overline{Q}$	"	26	6	4	25	"	"	"	"	"	"	"	"
T9	$3\overline{Q}$	"	28	8	29	9	"	"	"	"	"	"	"	"
T10	$4\overline{Q}$	"	13	14	12	32	"	"	"	"	"	"	"	"
T11	$5\overline{Q}$	"	35	15	16	17	"	"	"	"	"	"	"	"
T12	$6\overline{Q}$	"	39	18	40	19	"	"	"	"	"	"	"	"

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TABLE XVI. Threshold clock test.

Test condition		Parameter	Connect				Initial limits 25°C ±5°C ambient				Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	TB		TC			
							Min	Max	Min	Max		
U1	1Q	Threshold clock	3	24	2	21	440	585	900	1100	ns	20,22
U2	2Q	"	26	6	25	4	"	"	"	"	"	"
U3	3Q	"	28	8	9	29	"	"	"	"	"	"
U4	4Q	"	13	14	32	12	"	"	"	"	"	"
U5	5Q	"	35	15	17	16	"	"	"	"	"	"
U6	6Q	"	39	18	19	40	"	"	"	"	"	"

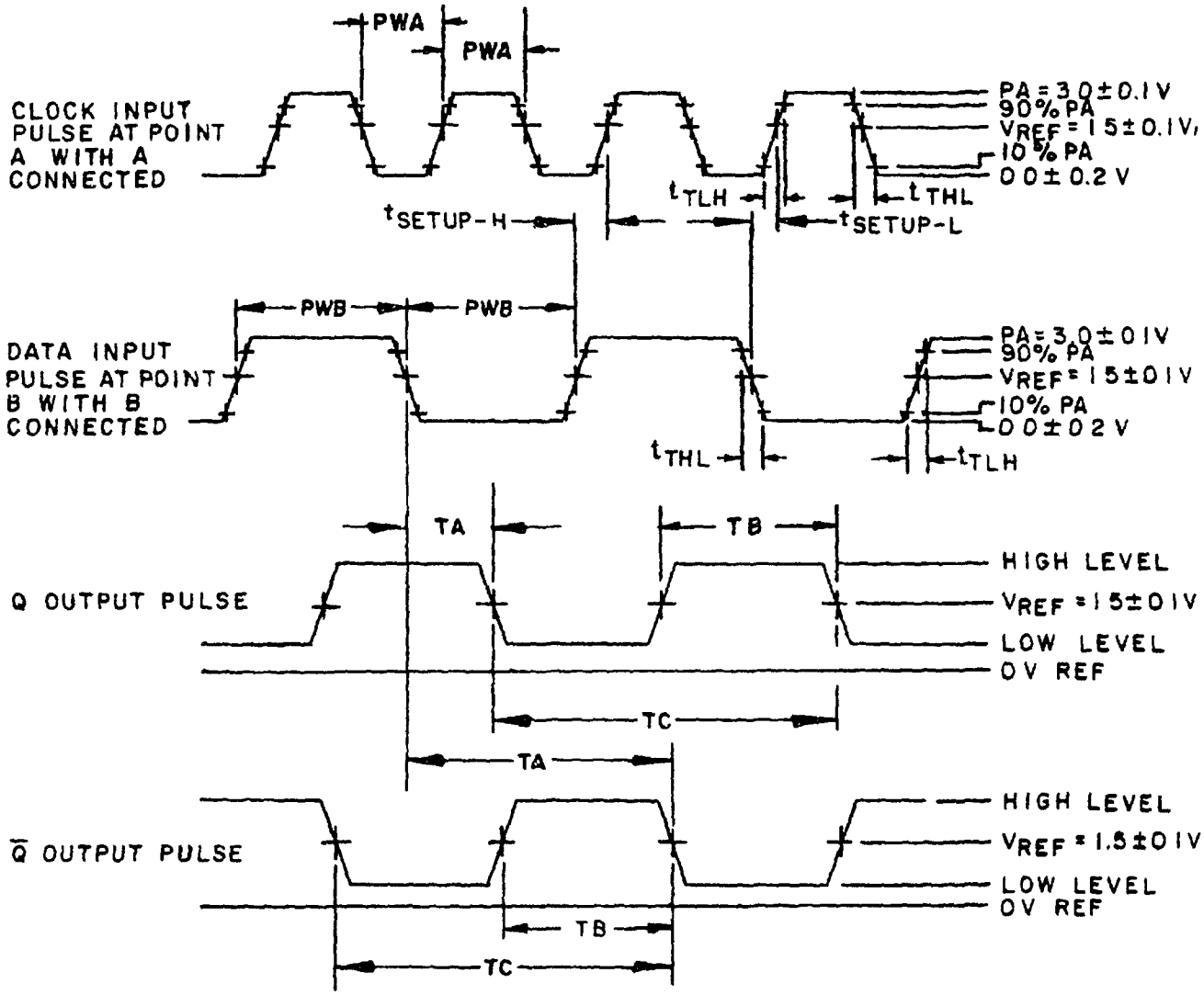
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NOTE: R1-R3 = 51Ω ±5%, 1 W.

* FIGURE 17. Setup time and hold time test circuit for table XIV.

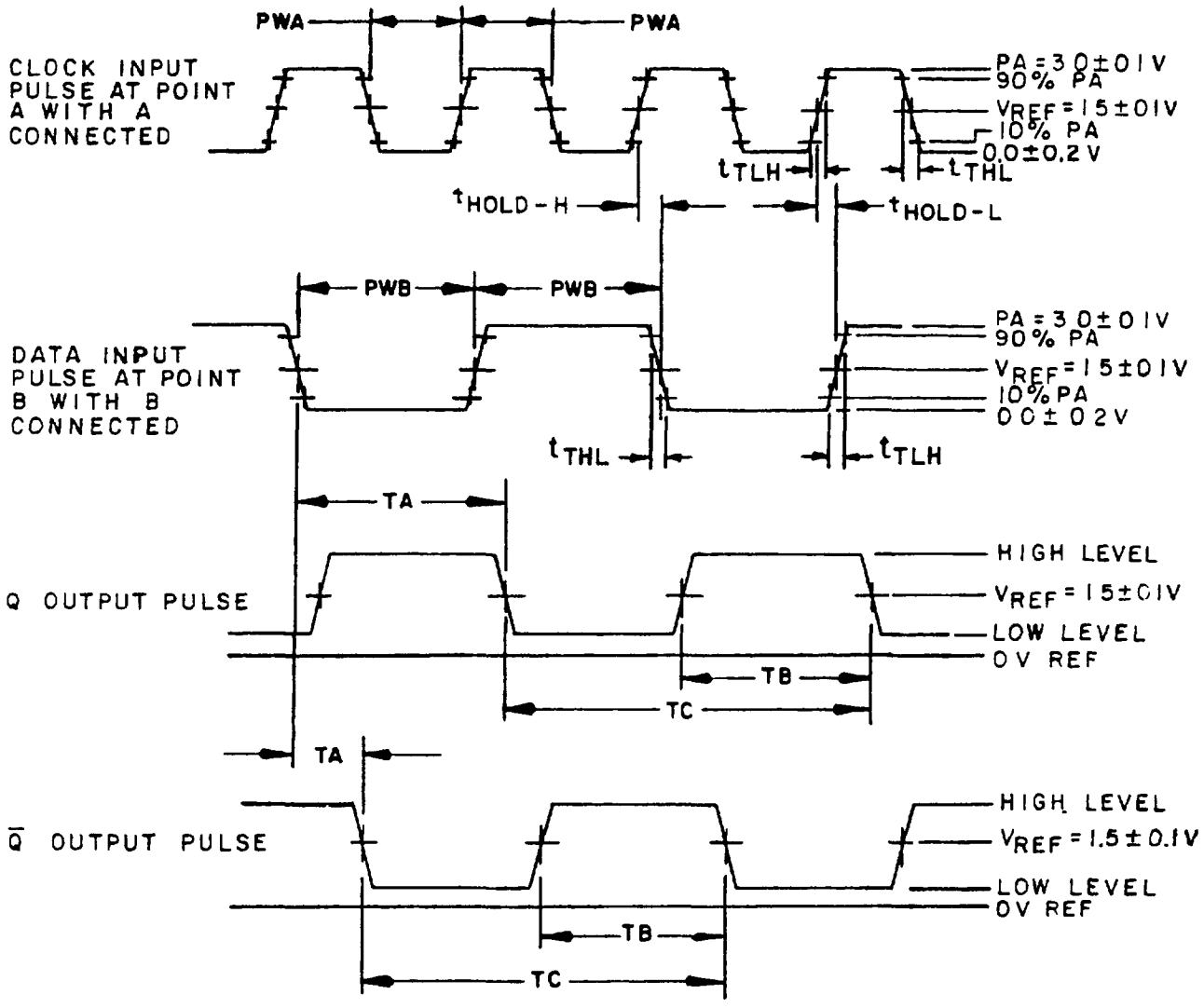
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Notes;
1. Input pulse parameters:
   PWA = 200 ns ±10%.
   PWB = 500 ns ±10%.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   $t_{SETUP-H} = t_{SETUP-L} = 20\ ^{+2}_{-0}$ ns.
2. The loads are not required.
3. Parameters TA, TB, and TC as measured on item D05 shall comply with electrical requirements of test conditions R1-R12 and shall be measured per waveforms above. Before making these measurements, the output levels shall be verified to be $\leq$ 0.4 V for the low level and $\geq$ 2.4 V of test conditions for the high level.

* FIGURE 18. Setup time test circuit waveforms for table XIV.

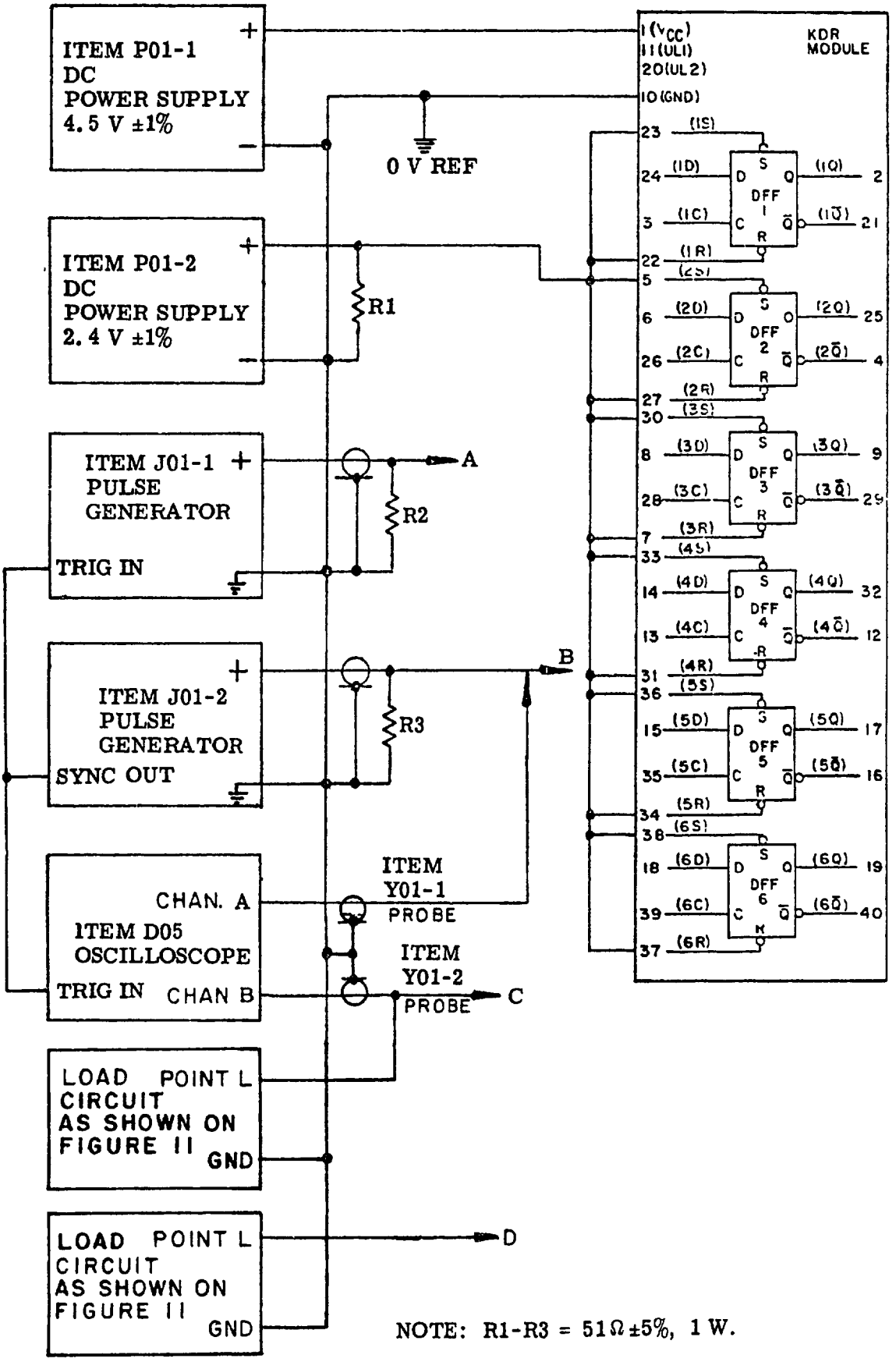
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Notes;
1. Input pulse parameters:
   PWA = 250 ns ±10%.
   PWB = 500 ns ±10%.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   $t_{HOLD-H} = t_{HOLD-L} = 5\ ^{+2}_{-0}$ ns.
2. The loads are not required.
3. Parameters TA, TB, and TC as measured on item D05 shall comply with electrical requirements of test conditions S1-S12 and shall be measured per waveforms above. Before making these measurements, the output levels shall be verified to be $\leq$ 0.4 V for the low level and $\geq$ 2.4 V for the high level.

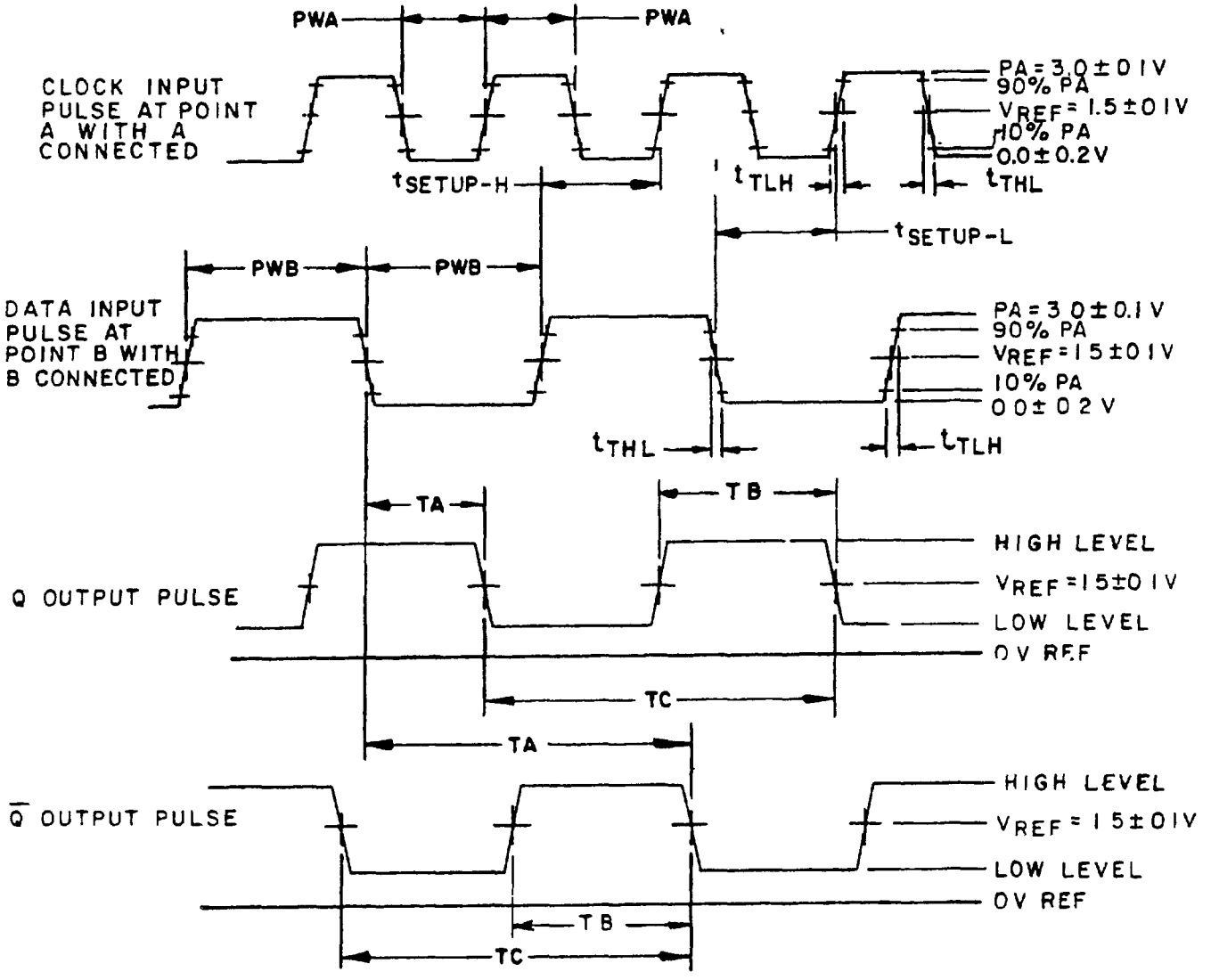
* FIGURE 19. Hold time test circuit waveforms for table XIV.

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* FIGURE 20. Maximum clock frequency/minimum clock pulse width and threshold clock test circuit for tables XV and XVI.

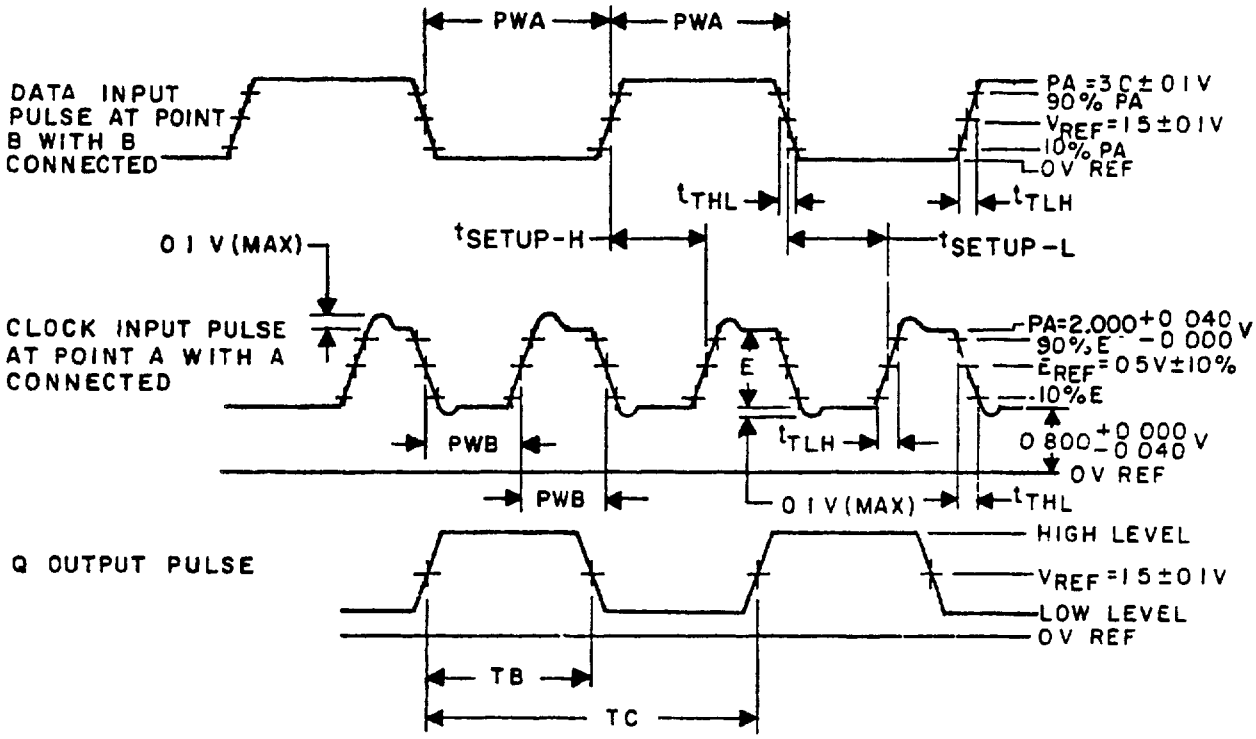
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Notes;
1. Input pulse parameters:
   PWA = 42 ±2 ns.
   PWB = 82 ±4 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   $t_{SETUP-H} = t_{SETUP-L}$ = 60 ns ±5%.
2. Both outputs of the circuit under test shall be loaded simultaneously.
3. Parameters TA, TB, and TC as measured on item D05 shall comply with electrical requirements of test conditions T1-T12 and shall be measured per waveforms above. Before making these measurements, the output levels shall be verified to be $\leq$ 0.4 V for the low level and $\geq$ 2.4 V for the high level.

* FIGURE 21. Maximum clock frequency/minimum clock pulse width test circuit waveforms for table XV.

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Notes:
1. Input pulse parameters:
   PWA = 500 ns ±10%.
   PWB = 250 ns ±10%.
   $t_{TLH} = t_{THL} = 25$ ns ±10%.
   $t_{SETUP-H} = t_{SETUP-L} = 125$ ns ±10%.
2. Both outputs of the circuit under test shall be loaded simultaneously.
3. Parameters TB and TC as measured on item D05 shall comply with electrical requirements of test conditions U1-U6 and shall be measured per waveforms above. Before making these measurements, the output levels shall be verified to be $\leq$ 0.4 V for the low level and $\geq$ 2.4 V for the high level.

* FIGURE 22. Threshold clock test circuit waveforms for table XVI.

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TABLE XVII. Propagation delay time (t_{PHL}) test.

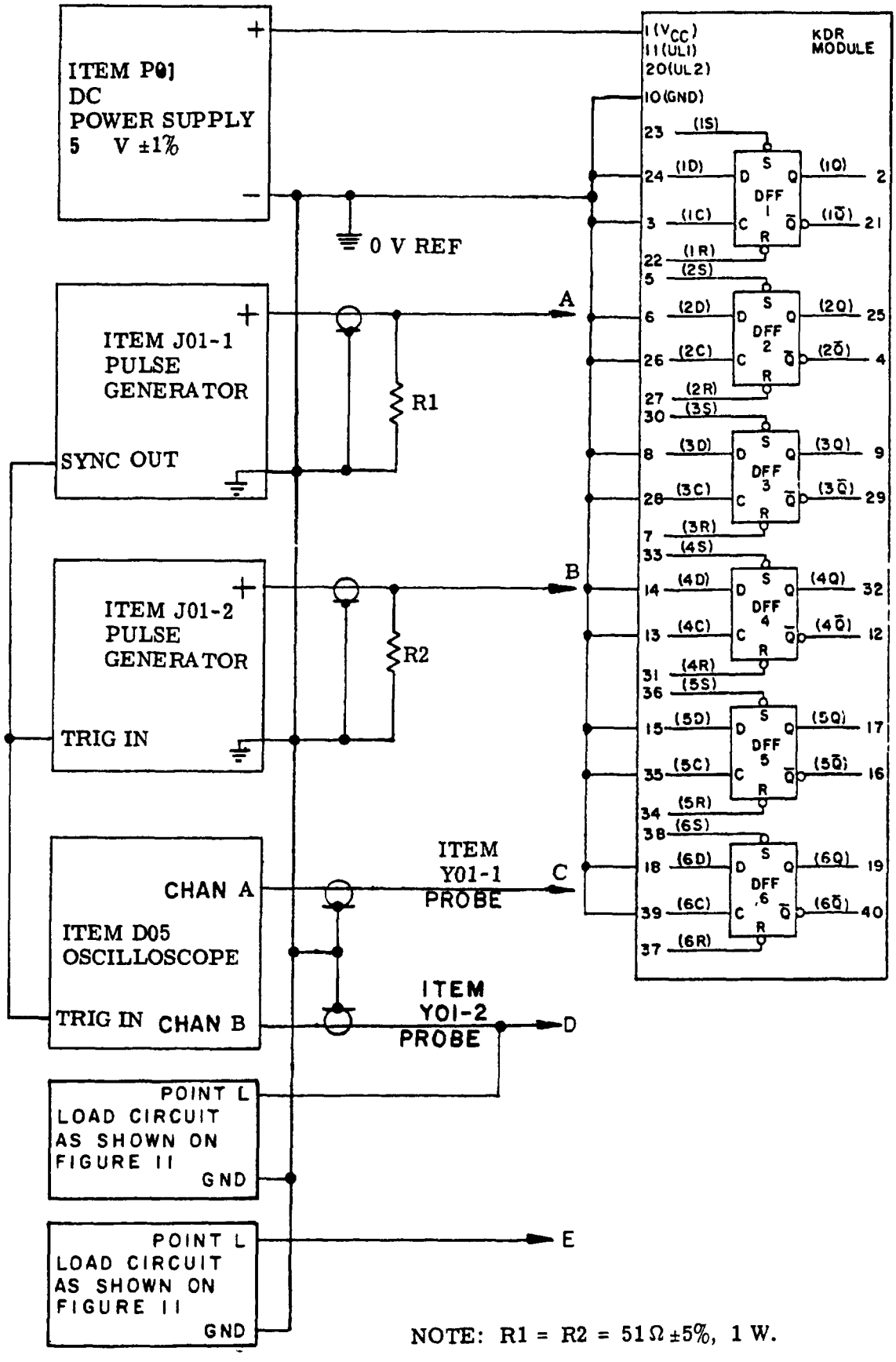
Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
V1	1Q Propagation delay time (t_{PHL}) (set- and reset-to-output)	23	22,C	B	2	21	6	45	ns	23,24
V2	1$\overline{Q}$ "	23,C	22	A	21	2	"	"	"	"
V3	2Q "	5	27,C	B	25	4	"	"	"	"
V4	2$\overline{Q}$ "	5,C	27	A	4	25	"	"	"	"
V5	3Q "	30	7,C	B	9	29	"	"	"	"
V6	3$\overline{Q}$ "	30,C	7	A	29	9	"	"	"	"
V7	4Q "	33	31,C	B	32	12	"	"	"	"
V8	4$\overline{Q}$ "	33,C	31	A	12	32	"	"	"	"
V9	5Q "	36	34,C	B	17	16	"	"	"	"
V10	5$\overline{Q}$ "	36,C	34	A	16	17	"	"	"	"
V11	6Q "	38	37,C	B	19	40	"	"	"	"
V12	6$\overline{Q}$ "	38,C	37	A	40	19	"	"	"	"

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TABLE XVIII. Propagation delay time (t_{PLH}) test.

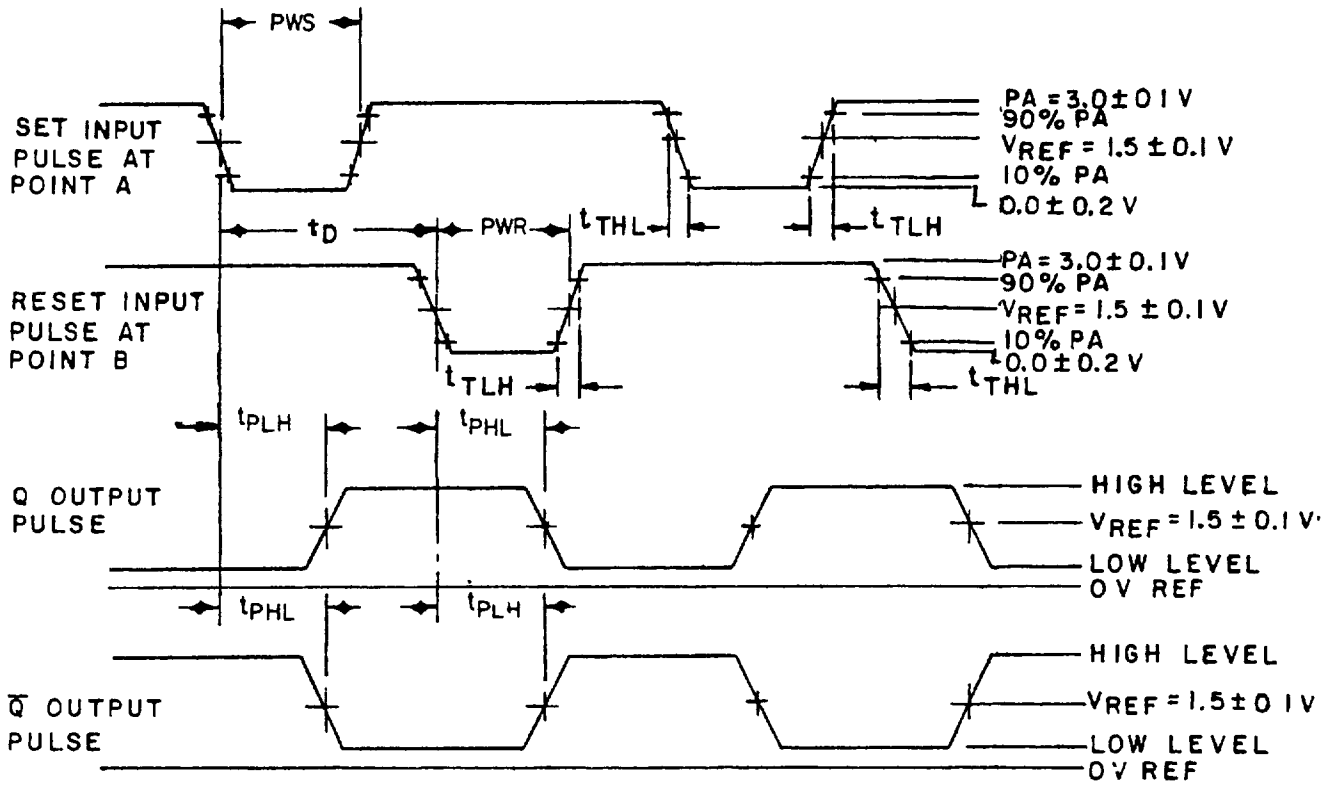
Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W1	1Q Propagation delay time (t_{PLH}) (set- and reset-to-output)	23,C	22	A	2	21	6	25	ns	23,24
W2	$1\overline{Q}$ "	23	22,C	B	21	2	"	"	"	"
W3	2Q "	5,C	27	A	25	4	"	"	"	"
W4	$2\overline{Q}$ "	5	27,C	B	4	25	"	"	"	"
W5	3Q "	30,C	7	A	9	29	"	"	"	"
W6	$3\overline{Q}$ "	30	7,C	B	29	9	"	"	"	"
W7	4Q "	33,C	31	A	32	12	"	"	"	"
W8	$4\overline{Q}$ "	33	31,C	B	12	32	"	"	"	"
W9	5Q "	36,C	34	A	17	16	"	"	"	"
W10	$5\overline{Q}$ "	36	34,C	B	16	17	"	"	"	"
W11	6Q "	38,C	37	A	19	40	"	"	"	"
W12	$6\overline{Q}$ "	38	37,C	B	40	19	"	"	"	"

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* **FIGURE 23.** Propagation delay time (t_{PHL} and t_{PLH}) test circuit for tables XVII and XVIII.

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NOTES:
Propagation delay time (t_{PHL}):
1. Both outputs of the circuit under test shall be loaded simultaneously.
2. Adjust item J01-1 and J02-2 to provide input pulse parameters as follows:
   PWS = PWR = 40 ±2 ns.
   PRR = 5 MHz ±5%.
   t_D = 100 ns ±5%.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
3. The values measured on item D05 shall comply with electrical requirements of test conditions V1-V12 and shall be measured per the above waveforms.
4. Before making the measurements, the output levels shall be verified to be $\leq$ 0.4 V for the low level and $\geq$ 2.4 V for the high level.

Propagation delay time (t_{PLH}):
1. Same as notes 1 and 2 above.
2. The values measured on item D05 shall comply with electrical requirements of test conditions W1-W12 and shall be measured per the above waveforms.
3. See note 4 above.

* FIGURE 24. Propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables XVII and XVIII.

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TABLE XIX. Output crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
X1	1Q Output crosstalk, low level	23,27,5,7,30,31,33,34,36,37,38	22	2	0	400	mV	25,26
X2	1$\overline{Q}$ Output crosstalk, high level	23,27,5,7,30,31,33,34,36,37,38	22	21	"	"	"	"
X3	1$\overline{Q}$ Output crosstalk, low level	22,27,5,7,30,31,33,34,36,37,38	23	21	"	"	"	"
X4	1Q Output crosstalk, high level	22,27,5,7,30,31,33,34,36,37,38	23	2	"	"	"	"
X5	2Q Output crosstalk, low level	22,23,5,7,30,31,33,34,36,37,38	27	25	"	"	"	"
X6	2$\overline{Q}$ Output crosstalk, high level	22,23,5,7,30,31,33,34,36,37,38	27	4	"	"	"	"
X7	2$\overline{Q}$ Output crosstalk, low level	22,23,27,7,30,31,33,34,36,37,38	5	4	"	"	"	"
X8	2Q Output crosstalk, high level	22,23,27,7,30,31,33,34,36,37,38	5	25	"	"	"	"
X9	3Q Output crosstalk, low level	22,23,27,5,30,31,33,34,36,37,38	7	9	"	"	"	"
X10	3$\overline{Q}$ Output crosstalk, high level	22,23,27,5,30,31,33,34,36,37,38	7	29	"	"	"	"
X11	3$\overline{Q}$ Output crosstalk, low level	22,23,27,5,7,31,33,34,36,37,38	30	29	"	"	"	"
X12	3Q Output crosstalk, high level	22,23,27,5,7,31,33,34,36,37,38	30	9	"	"	"	"
X13	4Q Output crosstalk, low level	22,23,27,5,7,30,33,34,36,37,38	31	32	"	"	"	"
X14	4$\overline{Q}$ Output crosstalk, high level	22,23,27,5,7,30,33,34,36,37,38	31	12	"	"	"	"
X15	4$\overline{Q}$ Output crosstalk, low level	22,23,27,5,7,30,31,34,36,37,38	33	12	"	"	"	"
X16	4Q Output crosstalk, high level	22,23,27,5,7,30,31,34,36,37,38	33	32	"	"	"	"
X17	5Q Output crosstalk, low level	22,23,27,5,7,30,31,33,36,37,38	34	17	"	"	"	"
X18	5$\overline{Q}$ Output crosstalk, high level	22,23,27,5,7,30,31,33,36,37,38	34	16	"	"	"	"
X19	5$\overline{Q}$ Output crosstalk, low level	22,23,27,5,7,30,31,33,34,37,38	36	16	"	"	"	"
X20	5Q Output crosstalk, high level	22,23,27,5,7,30,31,33,34,37,38	36	17	"	"	"	"

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TABLE XIX. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
X21	6Q Output crosstalk, low level	22,23,27,5,7,30,31,33,34,36,38	37	19	0	400	mV	25,26
X22	6$\overline{Q}$ Output crosstalk, high level	22,23,27,5,7,30,31,33,34,36,38	37	40	"	"	"	"
X23	6$\overline{Q}$ Output crosstalk, low level	22,23,27,5,7,30,31,33,34,36,37	38	40	"	"	"	"
X24	6Q Output crosstalk, high level	22,23,27,5,7,30,31,33,34,36,37	38	19	"	"	"	"

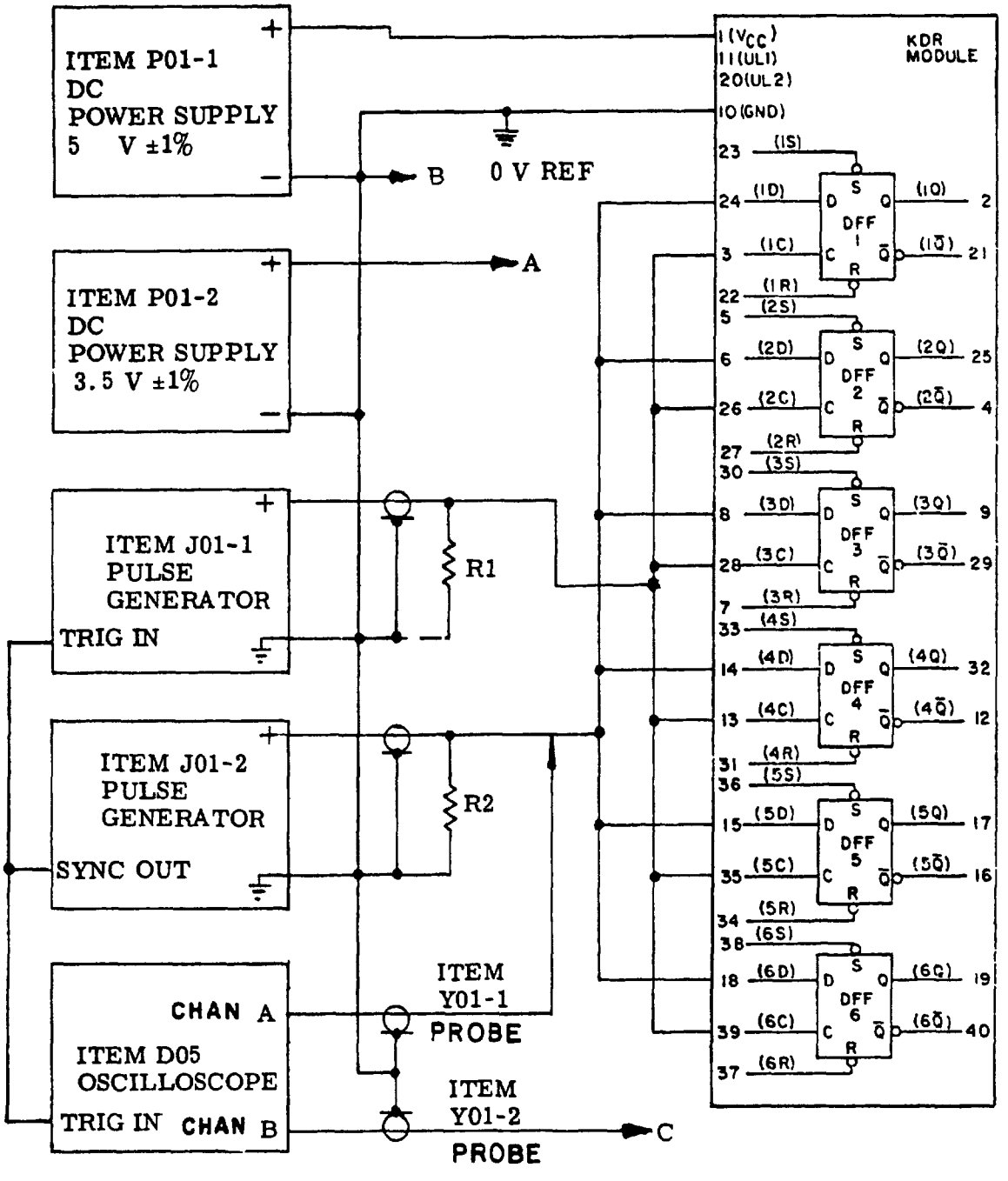
TABLE XX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
Y	Power supply transient amplitude	0	400	mV	27,28

TABLE XXI. Power supply transient rise time test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
Z	Power supply transient rise time	1	2	ms	29,30

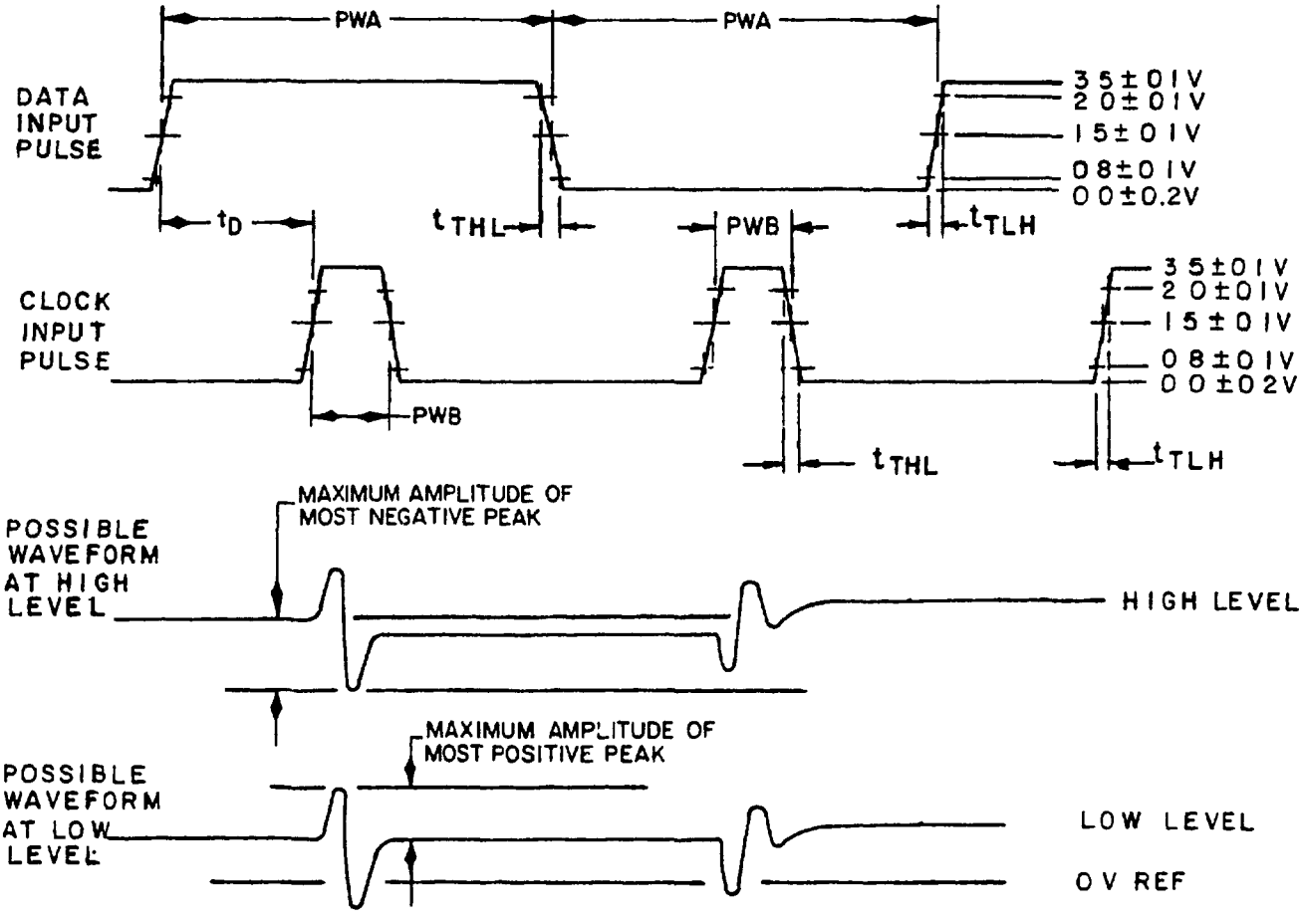
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NOTES:
1. R1=R2 = 51Ω ±5%, 1 W.
2. Load each output as specified on figure 11.

* FIGURE 25. Output crosstalk (low-level and high-level) test circuit for table XIX.

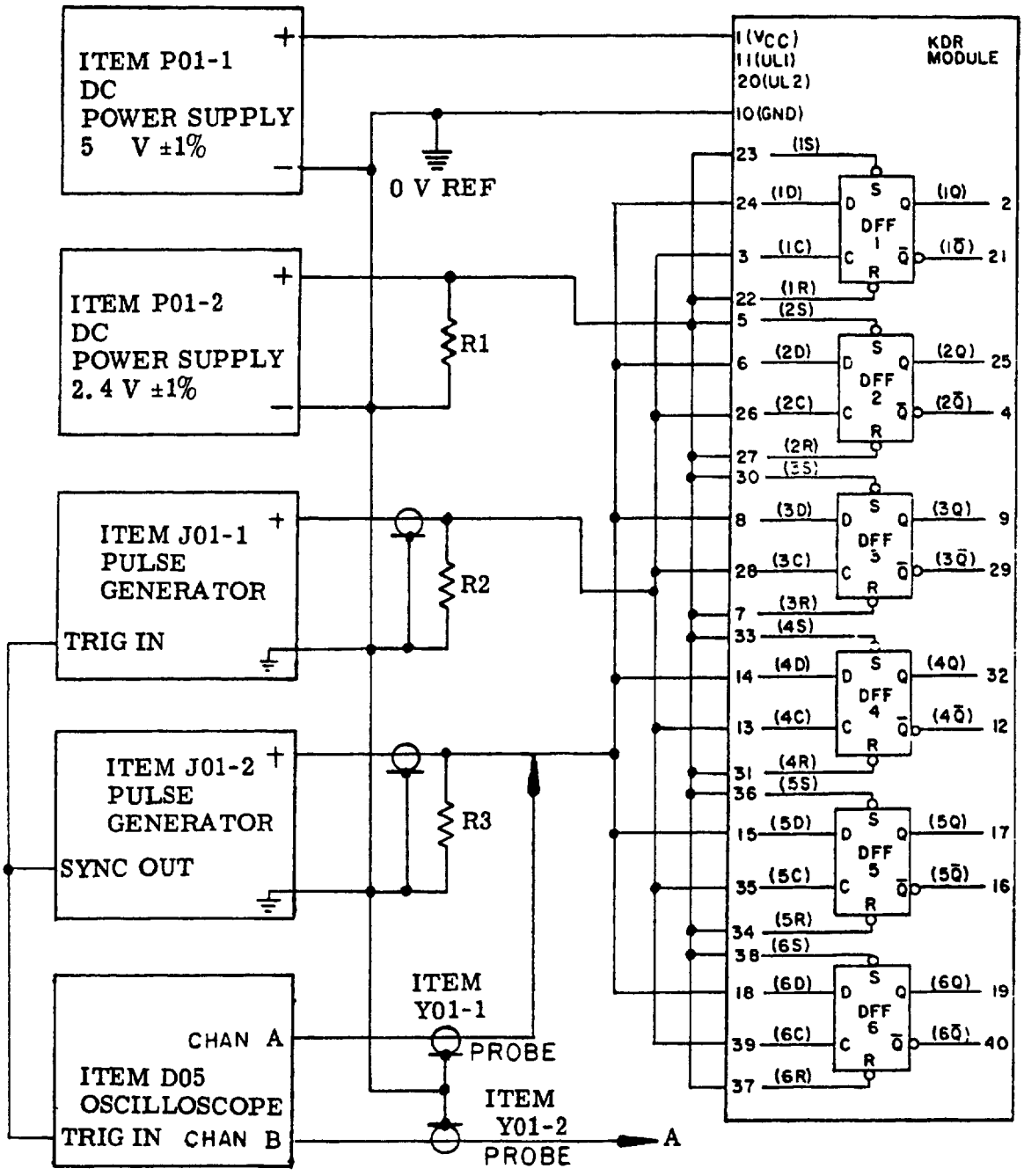
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NOTES: Input pulse parameters:
PWA = 500 ns ±10%.
PWB = 100 ns ±10%.
PRR (Clock) = 2 MHz ±10%.
$t_{TLH} = t_{THL}$ 3.5 ±1.5 ns (verified at each input).
t_D = 200 ns ±10%.

* FIGURE 26. Output crosstalk (low-level and high-level) test circuit waveforms for table XIX.

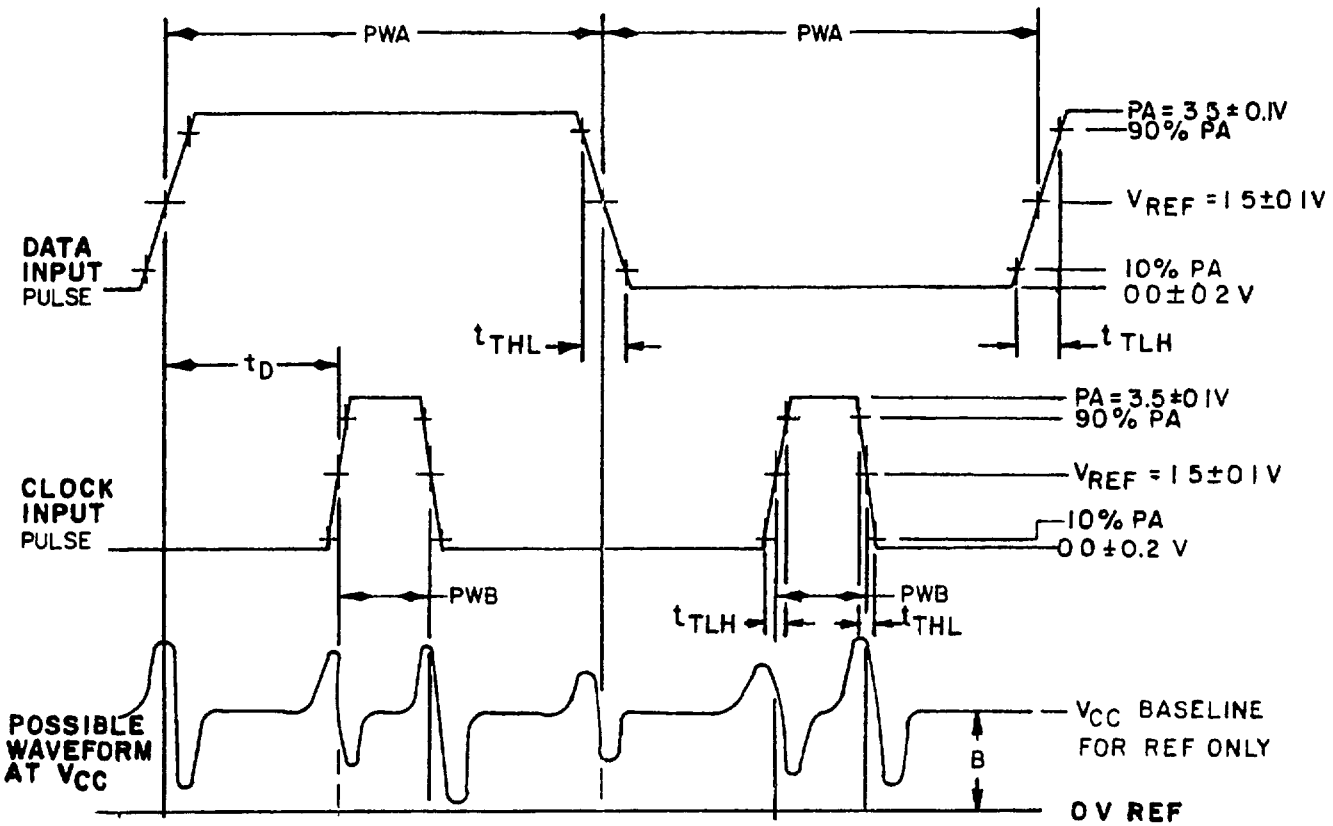
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NOTE: R1-R3 = 51Ω ±5%, 1 W.

* FIGURE 27. Power supply transient amplitude test circuit for table XX.

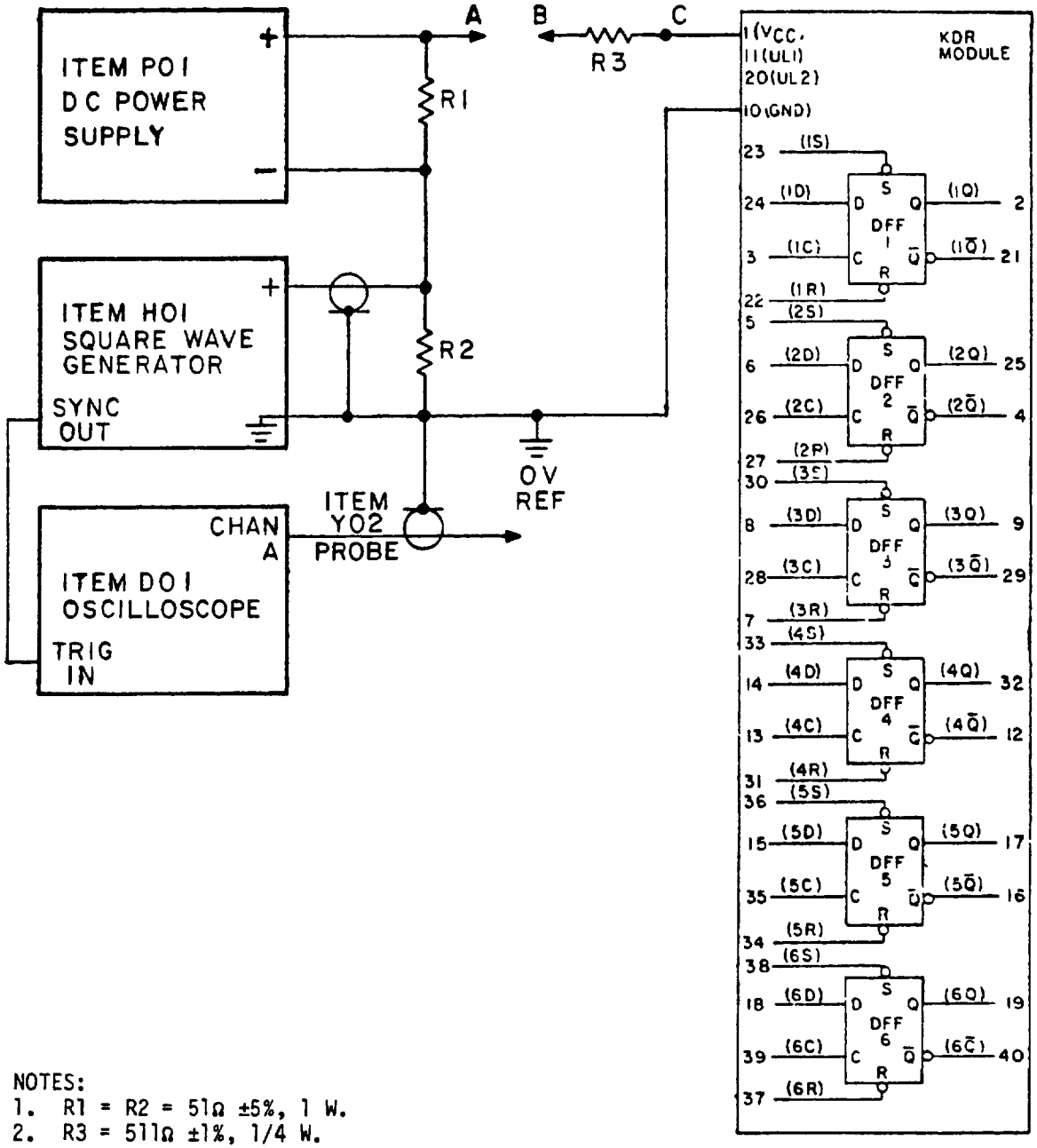
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NOTES:
1. Input pulse parameters:
   PWA = 500 ns ±10%.
   PWB = 100 ns ±10%.
   PRR (Clock) = 2 MHz ±10%.
   $t_{TLH} = t_{THL} = 10 \pm 1$ ns.
   t_D = 200 ns ±10%.
2. Connect point A as close as possible to pin 1 on the module. Observe power supply transient on item D05; typical waveforms are shown above.
3. B = measurement from the baseline to the most negative excursion of the waveform.

* FIGURE 28. Power supply transient amplitude test circuit waveforms for table XX.

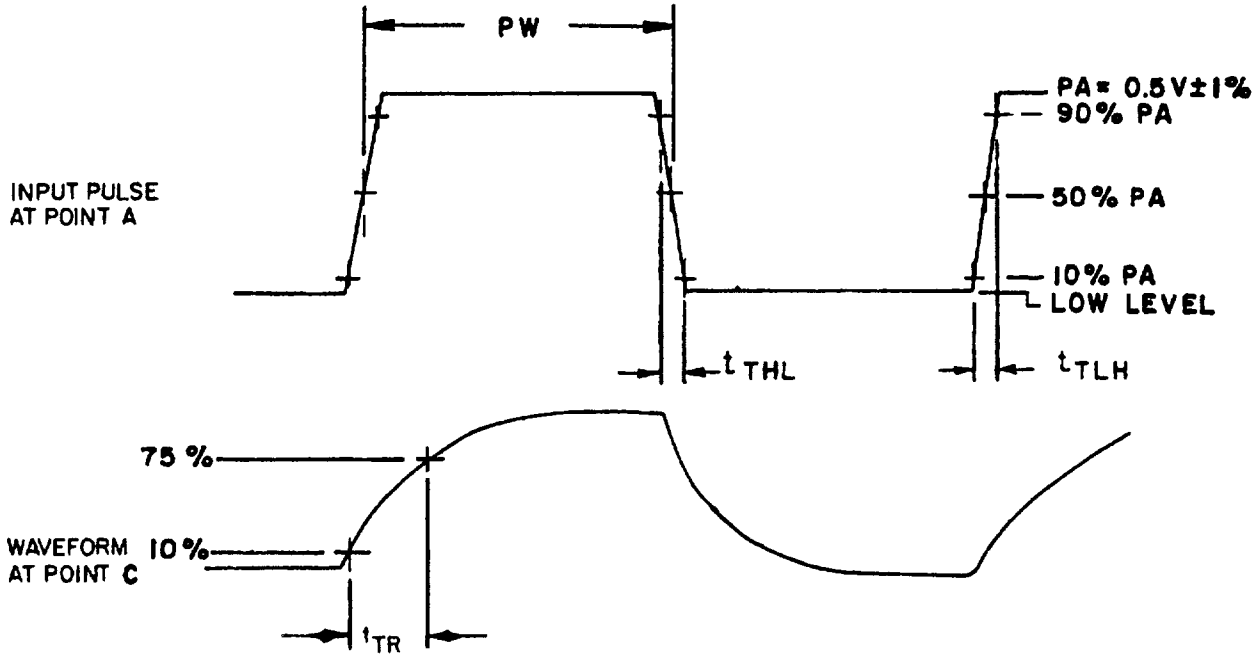
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. R3 = 511Ω ±1%, 1/4 W.

* FIGURE 29. Power supply transient rise time test circuit for table XXI.

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NOTES:
1. Input pulse parameters:
   PW = 10 ms ±10%.
   PRR = 50 Hz (reference).
   $t_{TLH} = t_{THL} = < 100$ ns (reference).
2. With point A open, adjust input as shown.
3. With point A open, adjust item P01 such that the low level of the input is 0.00 ±0.01 V.
4. Connect A and B; recheck low level and pulse parameters at point A.
5. Measure t_{TR} at point C.

* FIGURE 30. Power supply transient rise time test circuit waveforms for table XXI.

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